**Research Article**

# Induced Paths, Radius, and Extremal Neighborhood Independence in Graphs without 4-Cycles

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## Abstract

Let $M(n)$ be the maximum of $\ell(G) = S(G)/n$ over connected graphs of order $n$ with no 4-cycle subgraph and no induced path on six vertices, where $S(G)$ is the sum of $\alpha(G[N(v)])$. This maximum is $16/7$, $5/2$, $8/3$ and 3 at orders 7, 8, 9 and 10, and $2 + 2/n$ for every $n \geq 11$; for every $n \geq 7$ the extremal graph is unique, and $S \leq 3n$ with equality only for the Petersen graph. Every connected graph with no 4-cycle subgraph, of radius 4 and induced-path order 7, is $C_8$ or one graph in the 6-cycle extension classified in the text, and satisfies $S = 2n - b - m$, where $b$ is the number of vertices added on cycle edges and $m$ is the number of matching edges. For connected triangle-free graphs with no 4-cycle subgraph and no induced path on seven vertices, $2e \leq 3n$ with equality only for the Petersen graph, and the maximum of $2e$ on $n \geq 10$ vertices is $3n - \min\{n - 10,\ 5 + \mathbf{1}_{\{n \text{ even}\}}\}$; the extremal graphs are determined. Every connected graph with no 4-cycle subgraph and no induced path on seven vertices satisfies $S \leq 3n$, with equality only for the Petersen graph. The case in which the graph may contain a triangle uses a computer-assisted finite certificate. Consequently every connected graph with $\ell \leq 4$ satisfies Conjecture 133 of *Written on the Wall II*. The conjecture remains open for $\ell > 4$.

## 1 | Introduction

All graphs are finite and simple. For a graph $G$ with $n \geq 2$ vertices let $p(G)$ be the largest number of vertices of an induced path, $r(G)$ the radius, and

$$\ell(G) = \frac{1}{n} \sum_{v \in V(G)} \alpha\big(G[N(v)]\big),$$

where $\alpha$ is the independence number and $N(v)$ the open neighbourhood. Write $S(G)$ for the sum in the numerator, and write $c_{C_4}(G) = 1$ if $G$ contains no 4-cycle as a subgraph and $c_{C_4}(G) = 0$ otherwise. A graph with no 4-cycle subgraph is called *strong $C_4$-free*.

The contributions are the exact function $M(n)$ and the exact function $f(n)$ of Theorem 5.6, the radius-four classification, the bound $\mathrm{GEN}_3$, and REAL-133 for $\ell \leq 4$ in Corollary 5.10.

Theorem 3.6 determines $M(n)$, the maximum of $\ell$ over connected strong $C_4$-free $P_6$-free graphs of order $n \geq 7$, and proves that the extremal graph is unique. The value is $16/7$ at order 7, attained only by $\Gamma_0$; $5/2$ at order 8, attained only by the graph $H_8$ of Figure 1; $8/3$ at order 9, attained only by $P - v$; and 3 at order 10, attained only by the Petersen graph $P$. For every $n \geq 11$ the maximum is $2 + 2/n$, attained only by $\Gamma_{n-7}$. The same theorem gives $S \leq 3n$ for every order $n \geq 2$, with equality only for $P$. The change of extremal graph between orders 10 and 11 is read off the distance layers of an induced 6-cycle, drawn in Figure 2. At most three vertices lie in the first layer, on distinct antipodal pairs. If two or more of those

pairs are occupied, the second layer has size at most one, and the order is at most 10. At order 11 every graph in that branch has a single vertex on one antipodal pair and a second layer that induces a matching together with isolated vertices. If the matching has $m$ edges, then $S = 2n + 2 - m$, so equality $S = 2n + 2$ holds only when every vertex of the second layer is a leaf. The induced 5-cycle branch contributes at most $2n + 1$, and the chordal branch at most $2n - 2$. Below order 7 uniqueness can fail: the same division produces two extremal graphs at orders 4 and 6, namely $P_4$ and $K_{1,3}$ at order 4, both with $S = 6$, and $C_6$ together with $C_5$ plus one leaf at order 6, both with $S = 12$.
An enumeration of the connected strong $C_4$-free $P_6$-free graphs of order at most 12 agrees with the uniqueness statement for $n \geq 7$. It is not used in the proofs.

| $n$ | 7 | 8 | 9 | 10 | 11 | 12 |
|---|---|---|---|---|---|---|
| number of graphs | 43 | 87 | 183 | 355 | 699 | 1327 |
| maximum of $S$ | 16 | 20 | 24 | 30 | 24 | 26 |
| number of extremal graphs | 1 | 1 | 1 | 1 | 1 | 1 |

Theorem 4.3 classifies every connected strong $C_4$-free graph of radius 4 and induced-path order 7. The graph is $C_8$, or it is obtained from a 6-cycle, retaining the edges of the cycle, by nonempty sets $C_i$ of outer vertices, one set at each cycle vertex, with no edge between distinct sets $C_i$, each inducing a matching together with isolated vertices, together with at most one further vertex on each cycle edge, adjacent exactly to both endpoints of that edge. If $b$ is the number of vertices on cycle edges and $m$ is the number of matching edges, then $S = 2n - b - m$. In particular $\ell \leq 2$ throughout the class. The two skeletons that survive the list of 4-ciliates are $C_8$ and the 6-cycle with a pendant leaf at each vertex; $P_8$ has induced-path order 8, and the 4-cycle with pendant paths of length 2 contains a 4-cycle subgraph [10, 8]. The content of the theorem is the classification of the extensions and the count of $S$. Distance-four witnesses force the description: an edge with ends in distinct sets $C_i$ and $C_j$ completes an induced $P_8$. Figure 3(a) draws the path $w_5, v_5, v_0, x, y, v_2, v_3, w_3$. The path is chordless because the 6-cycle is induced, each off-cycle vertex meets only its designated cycle vertex, and the neighbourhoods of the two witnesses are constrained. Lemma 4.4 is the reverse construction: on any induced path of order at least three, every vertex off the 6-cycle is an endpoint.
Theorem 5.5 gives $2e \leq 3n$ for connected triangle-free strong $C_4$-free $P_7$-free graphs, with equality only for $P$. Theorem 5.6 determines the maximum $f(n)$ of $2e$ for every $n \geq 10$,

$$f(n) = 3n - \min\{n - 10,\ 5 + \mathbf{1}_{\{n \text{ even}\}}\},$$

together with the extremal graphs. The extremal family changes at the orders 15 and 16. Theorem 5.8 extends $S \leq 3n$, with equality only for $P$, from the triangle-free graphs to every connected strong $C_4$-free $P_7$-free graph. A minimal graph in the class that contains a triangle and satisfies $S \geq 3n$ has no cut vertex, satisfies $L(v) \geq 4$ at every vertex, and has no two adjacent vertices of degree two. Lemma 5.7 identifies the only graph with those local properties as the eleven-vertex subdivision $H_{3,2}$ of $K_5$, with $S = 29$.
Corollary 5.10 therefore gives REAL-133 for every connected graph with $\ell \leq 4$. The conjecture remains open for $\ell > 4$.

**Conjecture 1.1** (Graffiti/WOWII Conjecture 133)**.** *Every connected graph $G$ on at least two vertices satisfies*

$$r(G) + \ell(G)^{c_{C_4}(G)} \leq p(G).$$

The conjecture is number 133 in E. DeLaVina's list *Written on the Wall II* of conjectures of the program Graffiti.pc [7]. We follow the formal statement in the Formal Conjectures repository at the fixed commit 5faef0c4dc3d226f8f9a9fc8f581b6221283eb6e [11], the merge commit of pull request #5727 [12]. That commit removed the floor function from an earlier version of the file. The statement there reads `(rad : ℝ) + 1 G ^ cC4 ≤ (path G : ℝ)`. We work with this real-valued statement and call it REAL-133.
If $G$ contains a 4-cycle, the conjecture reads $p(G) \geq r(G) + 1$. A shortest path between two vertices at distance $\operatorname{diam}(G) \geq r(G)$ is induced, so the inequality holds. Only strong $C_4$-free graphs remain. Since $p$ and $r$ are integers, REAL-133 for such a graph is equivalent to

$$p(G) - r(G) \ \geq\ \lceil \ell(G) \rceil. \tag{1.1}$$

The comparisons with earlier structural work are recorded where they are used, as lemmas whose hypotheses can be checked without opening the citation, and they are not substitutes for the results above. Lemma 2.1 is the neighbourhood description in Lemma 2.1 of Chen [3]: in a strong $C_4$-free graph every neighbourhood is a matching together with isolated vertices, and $S = 2e - 3T$. The statements below are taken from the version https://doi.org/10.5281/zenodo.22054835.
His Lemma 7.18 applies to every graph with no 4-cycle subgraph and no induced $P_6$. On an induced 6-cycle, the neighbourhood of a vertex off the cycle is empty or exactly one antipodal pair. The class RES of his Definition 7.2 is not a hypothesis of that lemma. His Corollary 7.20 is in the same general setting: every vertex of the cycle is triangle-free, and its degree and the independence number of its neighbourhood are determined by which antipodal pairs are occupied. His Lemma 7.22

is the only one of these three results that assumes RES $\wedge$ $(C_6)$, and under that hypothesis it restricts the second distance layer.

After that general antipodal adjacency, the new content of Case 1 of Theorem 3.6 is the complete classification of the second layer. If two or more antipodal pairs are occupied, the second layer has at most one vertex, and the order in that branch is at most 10. This accounts for the value $S = 30$ of the Petersen graph at order 10, for the drop to $S = 24$ at order 11, and for uniqueness at every larger order. The exact function $M(n)$ is not a consequence of Lemmas 7.18, 7.20 and 7.22.

Remark 7.24 of the same preprint, on p. 23 of https://doi.org/10.5281/zenodo.22054835, records five admissible frames under the restricted hypothesis RES $\wedge$ $(C_6)$. The frames have orders $n = 6, 7, 8, 9$ and densities $\ell = 2$, $16/7$, $5/2$ and $8/3$. They are a finite check inside a class defined by radius 2, diameter 3 and the side condition (R3b), not a classification of every connected strong $C_4$-free $P_6$-free graph. The results outside that remark are the unrestricted class, the global description of the second distance layer, the exact function $M(n)$ at every order, the Petersen graph as the unique extremal graph of order 10, and $\Gamma_{n-7}$ as the unique extremal graph for every $n \geq 11$.

His Theorem 7.1 states that a counterexample of minimum order to his inequality (6.1) has radius at least 4. Inequality (6.1) asserts that every connected graph with no 4-cycle subgraph and with $\ell \geq 3$ satisfies $p \geq r + 3$. Theorem 2.4 gives $p \geq 2r - 1$, and $2r - 1 \geq r + 3$ for every integer $r \geq 4$. Thus $\ell \geq 3$ already yields $p \geq r + 3$ for every connected graph with no 4-cycle subgraph, by Theorem 7.1 together with Theorem 2.4. In the band $3 < \ell \leq 4$, Corollary 5.10 gives one further vertex for every connected strong $C_4$-free graph: $p \geq r + 4$. The real-valued formulation is what requires that further vertex: on this band the floor form $p \geq r + \lfloor \ell \rfloor$ asks only for $p \geq r + 3$.

His Theorem 6.7 produces an induced $P_6$ in a graph of diameter 2 with no 4-cycle subgraph and with $\ell > 3$. On the slice $r = 2$ and $3 < \ell \leq 4$ this is the same numerical conclusion as Theorem 5.9. Theorem 5.9 does not require the diameter to be 2, and Theorem 3.6 determines $M(n)$ at every order. Lemma 3.5 is the case $k = 6$ of Camby and Schaudt [2, Theorem 5], recorded as Lemma 2.8. It is used in Theorems 4.3 and 5.9. Theorem 3.6 uses the adjacency analysis after Lemma 3.4 together with Lemma 2.9, which is Theorem 3.2(d) of Karthick and Maffray [13]. The core list is Lemma 5.1, applied in Lemma 5.4 after Lemma 2.10. Remarks 3.7 and 3.8 record what those structure theorems do not imply.

Section 3 proves Theorem 3.6. Section 4 proves Theorem 4.3. Section 5 proves the triangle-free bound, the exact function $f(n)$, and Theorem 5.8. Section 6 applies these bounds to REAL-133 for $\ell \leq 4$, including the sharp density corollary for induced-path order 6, and records that the range $\ell > 4$ remains open. The supplement contains the rooted inequalities, the endpoint obstructions, the high-radius constructions, the one-triangle bound, the two-triangle classification, the edge-deficit estimates, and the candidate pairs in the ranges $4 < \ell \leq 5$ and $5 < \ell \leq 6$. Which of those arguments is a hypothesis of a theorem here is recorded once, in the dependency note at the beginning of the supplement. The finite-search certificate for Lemma 5.7 is of the form required here: a completeness argument in the text, together with the versioned checker `certificates/gen3/verify_gen3.py` at commit `dd555d305b4a69076cafb845087f00b4144f07cc`. From the repository root the command is `python3 certificates/gen3/verify_gen3.py`. Section A of the supplement records an independent execution of that command, the line it prints, and the five branch counts. An enumeration that does not meet this standard is not used as a hypothesis.

## 2 | Preliminaries

For $v \in V(G)$ let $d(v)$ be its degree and $t(v)$ the number of triangles containing $v$. Write $\lambda(v) = \lambda_G(v) = \alpha(G[N(v)])$, $S(G) = \sum_v \lambda(v)$, $e(G)$ for the number of edges and $T(G)$ for the number of triangles.

**Lemma 2.1.** *Let $G$ be strong $C_4$-free. Then:*

1. *two distinct vertices have at most one common neighbour;*
2. *for every $v$, $G[N(v)]$ is a matching together with isolated vertices, so $\lambda(v)$ equals the number of components of $G[N(v)]$ and $\lambda(v) = d(v) - t(v)$;*
3. *$S(G) = 2e(G) - 3T(G)$.*

*Proof.* (1) Two common neighbours $w, w'$ of $x, y$ give the 4-cycle $xwyw'$. (2) A vertex of $N(v)$ with two neighbours in $N(v)$ would have two common neighbours with $v$. So the components are single vertices and edges, each edge being a triangle at $v$. (3) Sum $d(v) - t(v)$ over $v$. □

The components of $G[N(v)]$ are the *units* of $v$. For a vertex set $U$, $G - U$ is the subgraph induced by $V \setminus U$, and $D_G(U) = S(G) - S(G - U)$ is the *deletion loss* of $U$. We write $L(v) = D_G(\{v\})$.

**Lemma 2.2.** *Let $G$ be strong $C_4$-free and $\emptyset \neq U \subseteq V(G)$. Then*

$$D_G(U) = \sum_{u \in U} \lambda(u) + \sum_{x \notin U} \#\{\text{units of } x \text{ contained in } U\}.$$

*In particular $L(v) = 2\lambda(v) - t(v) = 2d(v) - 3t(v)$.*

*Proof.* For $x \notin U$, deleting $U$ removes exactly the units of $x$ that lie inside $U$; a unit that is an edge with one end in $U$ survives as a vertex. For $U = \{v\}$: the $2t(v)$ neighbours of $v$ in triangles with $v$ keep their value, and each of the other $d(v) - 2t(v)$ neighbours loses one. The total is $\lambda(v) + d(v) - 2t(v) = 2\lambda(v) - t(v)$. □

**Lemma 2.3.** *Let $v$ be a cut vertex of $G$. Split the components of $G - v$ into two nonempty groups, and let $G_1, G_2$ be the subgraphs induced by the two groups together with $v$. Then $S(G) = S(G_1) + S(G_2)$.*

*Proof.* $G[N(v)]$ has no edges between the groups, so $\lambda_G(v) = \lambda_{G_1}(v) + \lambda_{G_2}(v)$. Every other vertex keeps its neighbourhood. □

We use three published results.

**Theorem 2.4** (Erdős, Saks and Sós [9])**.** *Every connected graph $G$ has an induced path on at least $2r(G) - 1$ vertices.*

**Theorem 2.5** (Camby and Schaudt [2, Theorem 5])**.** *Let $k \geq 4$ and let $G$ be a connected graph with no induced path on $k$ vertices. Then $G$ has a connected dominating set $X$ such that $G[X]$ has no induced path on $k-2$ vertices or $G[X]$ is isomorphic to the cycle $C_k$.*

**Theorem 2.6** (Seinsche [14])**.** *If a graph on at least two vertices has no induced path on four vertices, then it or its complement is disconnected.*

Theorem 2.5 is the implication (i)⇒(ii) of [2, Theorem 5] applied with $H = G$. We write $P_k$ for the path on $k$ vertices, and "$G$ is $P_k$-free" means that $G$ has no induced $P_k$, that is, $p(G) \leq k - 1$. A connected graph of radius $r$ has at least $2r$ vertices.

**Lemma 2.7.** *Let $H$ be a connected strong $C_4$-free $P_4$-free graph on at least two vertices. Then $H$ has a vertex adjacent to all other vertices.*

*Proof.* By Theorem 2.6, $H$ is the join of two nonempty vertex sets $A$ and $B$. If $|A|, |B| \geq 2$, then $a, a' \in A$ and $b, b' \in B$ span the 4-cycle $aba'b'$. So one side is a single vertex, and it is adjacent to everything. □

**Lemma 2.8.** *Let $G$ be a connected graph with no induced path on six vertices. Then $G$ has a connected dominating set $X$ such that $G[X]$ has no induced path on four vertices, or else $G[X]$ is isomorphic to $C_6$.*

This is Theorem 2.5 at $k = 6$. The hypotheses of that theorem are $k \geq 4$ and connectedness of a graph with no induced path on $k$ vertices, both of which hold, and the conclusion is the implication (i)⇒(ii) of [2, Theorem 5] applied with $H = G$. Here $k - 2 = 4$ and $C_k = C_6$. Lemma 3.5 uses this conclusion and no other part of the cited theorem. Theorem 3.6 does not use Lemma 3.5. Its arguments are the adjacency analysis of an induced 6-cycle, in the form recorded after Lemma 3.4, and Lemma 2.9 in the induced 5-cycle branch. Lemma 3.5 is used in the proof of Theorem 4.3 and in the case $r = 3$ of Theorem 5.9. Strong $C_4$-freeness is not required for the dominating set; it enters only in Lemma 3.5.

**Lemma 2.9** (Karthick and Maffray [13, Theorem 3.2(d)])**.** *Let $G$ be a graph with no induced $P_6$, no induced $C_4$ and no induced $C_6$, and let $C = c_0c_1c_2c_3c_4c_0$ be an induced 5-cycle, indices modulo 5. For each $i$ let $W_i$ be the set of vertices outside $C$ whose neighbourhood on $C$ is exactly $\{c_i\}$, and let $X_{i,i+1}$ be the set of vertices outside $C$ whose neighbourhood on $C$ is exactly $\{c_i, c_{i+1}\}$. Then, for every $i$, at least one of $X_{i,i+1}$ and $X_{i+1,i+2}$ is empty, at least one of $W_i$ and $W_{i+2}$ is empty, and at least one of $X_{i,i+1}$ and $W_{i-1} \cup W_{i+2}$ is empty.*

In [13, Theorem 3.2] the cycle is labelled $v_1, \ldots, v_5$ and $C_4$ denotes an induced cycle. The translation $v_{i+1} = c_i$ carries their sets onto $W_i$ and $X_{i,i+1}$ as defined here, and the three emptiness assertions are the whole of their part (d). A strong $C_4$-free graph has no induced $C_4$, so it meets the induced-cycle hypothesis whenever it is $P_6$-free. Case 2 of Theorem 3.6 uses the first two assertions. The third is included so that the quoted part is complete.

**Lemma 2.10.** *Let $G$ be a connected triangle-free strong $C_4$-free $P_7$-free graph with no cut vertex. Then $G$ has no clique cutset.*

*Proof.* The graph is triangle-free, so every clique has order at most two, and there is no clique cutset of order one. Suppose $\{u, v\}$ is a clique cutset. Then $uv$ is an edge. Neither vertex is a cut vertex, so every component of $G - \{u, v\}$ is adjacent to both $u$ and $v$, and there are at least two such components. A shortest $u$–$v$ path through one component has at least four edges: one internal vertex would form a triangle with $uv$, and two internal vertices would form a 4-cycle with $uv$. On each of two such paths, let $x, y, z$ be the three vertices immediately preceding $v$, in that order from the $u$-side, and

write $x', y', z'$ for the same choice in the second component. The path $x, y, z, v, z', y', x'$ is an induced $P_7$. Each side is a subpath of a shortest path, so it has no chord, and there is no edge between $\{x, y, z\}$ and $\{x', y', z'\}$ because those sets lie in different components of $G - \{u, v\}$. This contradicts $p(G) \le 6$. □

## 3 | Small-path mass bounds

**Lemma 3.1.** *If $G$ is connected and strong $C_4$-free with $r(G) = 1$, then $S(G) = 2(n-1) - m$, where $m$ is the number of edges of $G[N(u)]$ for a dominating vertex $u$. Hence $\ell(G) < 2$, and (1.1) holds.*

*Proof.* Let $v \neq u$. By Lemma 2.1(1), $v$ has at most one neighbour other than $u$, and such a neighbour lies in $N(u)$. So $G[N(v)]$ is a clique and $\lambda(v) = 1$; also $\lambda(u) = (n-1) - m$. Hence $S = 2(n-1) - m < 2n$ and $\lceil \ell \rceil \le 2$. If $G$ is complete then $G \in \{K_2, K_3\}$, $\ell = 1$ and $p = 2 = r + 1$. Otherwise two nonadjacent vertices of $N(u)$ together with $u$ form an induced $P_3$, so $p \ge 3 \ge r + \lceil \ell \rceil$. □

**Lemma 3.2.** *If $G$ is connected and strong $C_4$-free with $r(G) \ge 2$, then $p(G) \ge 4$.*

*Proof.* If $G$ were $P_4$-free, Lemma 2.7 would give a dominating vertex, so $r(G) \le 1$. □

**Lemma 3.3.** *If $G$ is connected, strong $C_4$-free and $P_5$-free, then $S(G) \le 2n - h$, where $h$ is the number of vertices $v$ with $\lambda(v) \ge 3$. In particular $S(G) \le 2n$.*

*Proof.* Call $v$ *simplicial* if $\lambda(v) = 1$ and *high* if $\lambda(v) \ge 3$. For $x \in N(v)$ put $X_x = N(x) \setminus N[v]$; then $N(x) \subseteq \{v\} \cup \{\bar{x}\} \cup X_x$, where $\bar{x}$ is the partner of $x$ if $x$ lies in an edge unit of $v$.
*Key fact.* Let $x, y$ lie in different units of $v$. A common vertex of $X_x$ and $X_y$ would give the 4-cycle $vxwy$, so $X_x \cap X_y = \emptyset$. If $x' \in X_x$ and $y' \in X_y$ are nonadjacent, then $x'xvyy'$ is an induced $P_5$: $x \not\sim y$ because they lie in different units, $x \not\sim y'$ and $y \not\sim x'$ by disjointness, and $v$ is adjacent to neither $x'$ nor $y'$. So $X_x$ is complete to $X_y$. If both are nonempty and $|X_x| \ge 2$, a vertex of $X_y$ has two neighbours in $X_x$, which are also neighbours of $x$, a contradiction to Lemma 2.1(1). Hence if both are nonempty, then $|X_x| = |X_y| = 1$.
*Main step.* For each high $v$ we find $\lambda(v) - 1$ simplicial vertices whose only high neighbour is $v$, one in each of $\lambda(v) - 1$ distinct units of $v$.
Case 1: some $u \in N(v)$ has $|X_u| \ge 2$. By the key fact $X_y = \emptyset$ for every $y$ outside the unit of $u$. Then $N(y) \subseteq \{v, \bar{y}\}$, which is a clique, so $y$ is simplicial; so is $\bar{y}$. Hence $v$ is the only high neighbour of $y$, and the $\lambda(v) - 1$ units other than that of $u$ supply the vertices.
Case 2: $|X_x| \le 1$ for all $x \in N(v)$. Then $\lambda(x) \le 2$ for $x \in N(v)$. Call a unit *active* if one of its vertices $x$ has $X_x \neq \emptyset$. Suppose two units are active, with $y' \in X_y$ and $z' \in X_z$, and let $x$ lie in a third unit. By the key fact $y' \sim z'$, and $xvyy'z'$ is an induced $P_5$: $x \not\sim y$ (different units), $x \not\sim y', z'$ (disjointness), $v \not\sim y', z'$, and $y \not\sim z'$ (disjointness). As $\lambda(v) \ge 3$, a third unit exists, so at most one unit is active. Every vertex of an inactive unit is simplicial and its only high neighbour is $v$. This gives at least $\lambda(v) - 1$ units.
*Counting.* The chosen vertices for distinct high vertices are distinct, since each has a unique high neighbour. So $\sum_{v \text{ high}} (\lambda(v) - 1) \le \#\{\text{simplicial vertices}\}$, and

$$S = \sum_v \lambda(v) \le 2n - \#\{\text{simplicial}\} + \sum_{v \text{ high}} (\lambda(v) - 2) \le 2n - h.$$

□

**Lemma 3.4.** *Let $G$ be connected, strong $C_4$-free and $P_6$-free with a dominating induced cycle $C = v_0 v_1 \cdots v_5$, and let $W = V(G) \setminus V(C)$. Then every $w \in W$ is adjacent to exactly one antipodal pair $\{v_i, v_{i+3}\}$ and to no other vertex of $C$, distinct vertices of $W$ use distinct pairs, and $W$ is independent. Hence $|W| \le 3$, $S(G) = 12 + 4|W| < 3n$, and $r(G) \le 2$ if $W \neq \emptyset$.*

*Proof.* Indices are modulo 6. Let $w \in W$; it has a neighbour on $C$. If $w$ has exactly one neighbour $v_i$, then $wv_iv_{i+1}v_{i+2}v_{i+3}v_{i+4}$ is an induced $P_6$. If $N(w) \cap C = \{v_i, v_{i+1}\}$, then $wv_{i+1}v_{i+2}v_{i+3}v_{i+4}v_{i+5}$ is an induced $P_6$. Two neighbours at distance 2 on $C$ give a 4-cycle. A set of at least three vertices of $C_6$ contains two vertices at distance 2. So $N(w) \cap C$ is an antipodal pair. Two vertices of $W$ with the same pair give two common neighbours. If $w \sim w'$ with distinct pairs, some $v_i \in N(w)$ and $v_j \in N(w')$ are adjacent on $C$, and $wv_iv_jw'$ is a 4-cycle. There are no triangles, $e(G) = 6 + 2|W|$, and Lemma 2.1(3) gives $S = 12 + 4|W|$. Finally, if $w$ uses $\{v_i, v_{i+3}\}$, then every vertex of $C$ is within distance 2 of $v_i$ (the vertex $v_{i+3}$ through $w$), and each $w' \in W$ has a neighbour in $\{v_{i-1}, v_i, v_{i+1}\}$, because each of the three antipodal pairs meets this set. So $v_i$ has eccentricity 2. □

The same trace analysis applies to any vertex adjacent to an induced 6-cycle in a connected strong $C_4$-free $P_6$-free graph, whether or not the cycle dominates. Theorem 3.6 uses that form.

**Lemma 3.5.** *Let $G$ be connected, strong $C_4$-free and $P_6$-free. Then either $G$ has a vertex of eccentricity at most 2, or $G = C_6$.*

*Proof.* Lemma 2.8 supplies a connected dominating set $X$. If $G[X] \cong C_6$, Lemma 3.4 gives $G = C_6$ or a vertex of eccentricity 2. Otherwise $G[X]$ is $P_4$-free. If $|X| = 1$, the vertex in $X$ dominates $G$. If $|X| \geq 2$, Lemma 2.7 gives $u \in X$ adjacent to all of $X \setminus \{u\}$, and since $X$ dominates $G$, every vertex is within distance 2 of $u$. □

**Theorem 3.6.** *Let $M(n)$ be the maximum of $\ell(G)$ over connected strong $C_4$-free $P_6$-free graphs of order $n \geq 7$. Then*

$$M(n) = \begin{cases} 16/7 & \text{if } n = 7, \\ 5/2 & \text{if } n = 8, \\ 8/3 & \text{if } n = 9, \\ 3 & \text{if } n = 10, \\ 2 + 2/n & \text{if } n \geq 11. \end{cases}$$

*For every $n \geq 7$ the extremal graph is unique up to isomorphism. For $n = 10$ it is the Petersen graph $P$. For $n = 9$ it is $P - v$. For $n = 8$ it is the graph $H_8$ of Figure 1. For $n = 7$ and for every $n \geq 11$ it is the graph $\Gamma_{n-7}$ of that figure: a 6-cycle, a vertex $a$ adjacent exactly to one antipodal pair, and $n - 7$ leaves adjacent only to $a$. In particular, every connected strong $C_4$-free $P_6$-free graph $G$ of order $n \geq 2$ satisfies $S(G) \leq 3n$, with equality if and only if $G$ is the Petersen graph. At orders 4 and 6 the maximum is attained by two nonisomorphic graphs.*

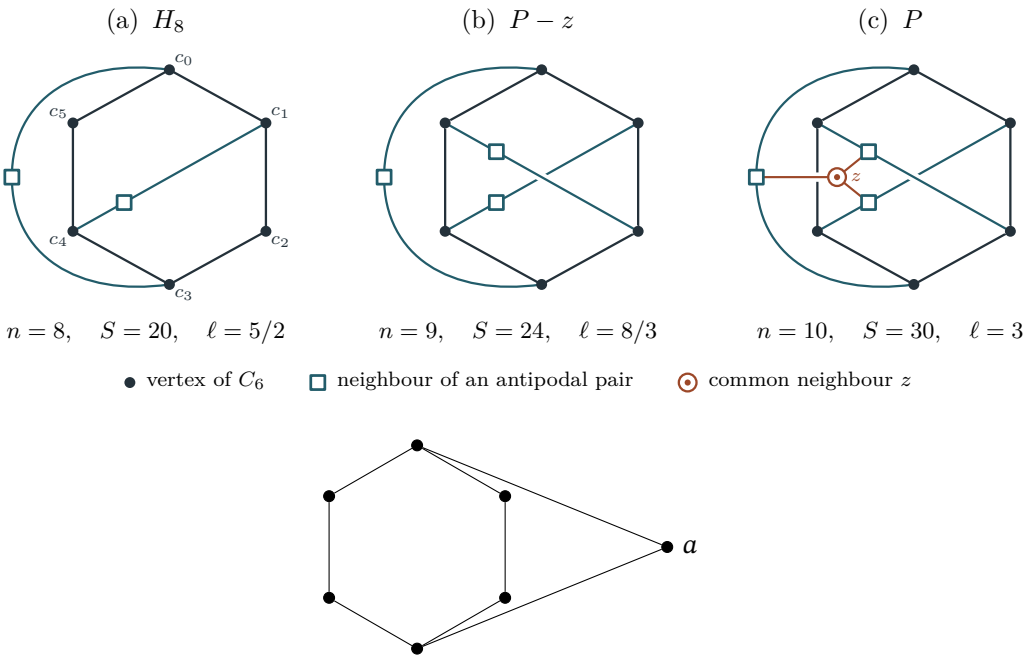



**FIGURE 1** | Extremal graphs for $M(n)$. In $H_8$, each of two antipodal pairs on the 6-cycle has its own common neighbour outside the cycle. Adding a common neighbour of the third pair gives $P - z$. Joining a new vertex $z$ to the three outside vertices gives the Petersen graph $P$. The three graphs are triangle-free, so $S = 2e$. The drawing underneath is $\Gamma_0$, the unique graph of order 7 with $\ell = 16/7$: the marked vertex $a$ is adjacent to one antipodal pair. Attaching any number of leaves at $a$ gives the unique extremal graph of each order $n \geq 11$, with $S = 2n + 2$.

*Proof.* Throughout, $S = 2e - 3T$ by Lemma 2.1. A chordless cycle of length at least seven contains an induced $P_6$. Thus the only possible holes have lengths five and six.

*Case 1: an induced six-cycle.* Label its vertices $c_0, \ldots, c_5$, indices modulo 6, and write $A$ for the set of vertices at distance one from the cycle. The trace argument in the proof of Lemma 3.4 shows that each $a \in A$ meets the cycle in one antipodal pair. Distinct vertices of $A$ use distinct pairs, and $A$ is an independent set: an edge of $A$ joins distinct pairs, and some contact of one end is adjacent on the cycle to some contact of the other, giving a 4-cycle. Hence $k = |A| \leq 3$. Figure 1 shows the three extremal graphs of this branch, and Figure 2 shows the two distance layers used in the count.

Every vertex $x$ at distance two from the cycle is adjacent to every vertex of $A$. Such an $x$ has at least one neighbour in $A$. If $x \sim a$ and $x \nsim b$ for $a, b \in A$, choose contacts so that $c_i \in N(a)$ and $c_{i+2} \in N(b)$. This is possible: the pairs are distinct, so

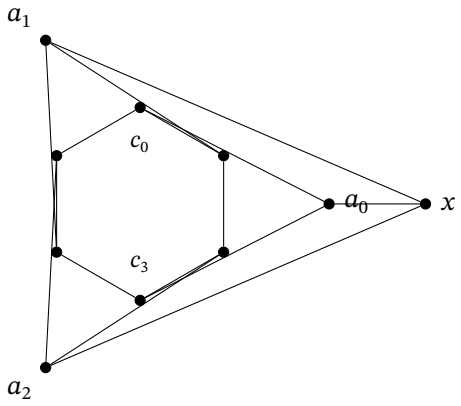



**FIGURE 2** | Distance layers of an induced 6-cycle in Case 1 of Theorem 3.6. Each vertex $a_i$ of the first layer meets the cycle in one antipodal pair, the three pairs are distinct, and the first layer is an independent set, so its size is $k \leq 3$. A vertex $x$ of the second layer is adjacent to every vertex of the first layer and to no vertex of the cycle. If $k \in \{2, 3\}$, a second vertex of the second layer would have two common neighbours in the first layer, so that layer has size at most one and the order is at most 10. There is no third layer.

some contact of $b$ lies at cycle distance two from some contact of $a$, and we orient the cycle from that contact of $a$ toward that contact of $b$. The path $x, a, c_i, c_{i+1}, c_{i+2}, b$ is then an induced $P_6$. Indeed $x$ meets the cycle in nothing, $a$ meets the cycle only in its antipodal pair (neither $c_{i+1}$ nor $c_{i+2}$), $b$ meets the cycle only in its antipodal pair (so $b \sim c_{i+2}$ and $b \nsim c_i, c_{i+1}$), and $a \nsim b$.

There is no vertex at distance three or more from the cycle. If there were, the cycle would have a vertex $z$ at distance exactly three. A shortest $z$–cycle path is then a four-vertex geodesic $z, x, a, c_i$, with $x$ at distance two and $a$ at distance one. Extending that geodesic by $c_{i+1}, c_{i+2}$ gives an induced $P_6$: the vertex $z$ has no neighbour in $A$ or on the cycle, and $x$ has no neighbour on the cycle.

Let $h$ be the number of vertices at distance two. If $k = 0$, then $A$ is empty, so the cycle has no neighbour and $G = C_6$. If $k = 1$, every distance-two vertex is adjacent to the unique vertex $a \in A$. The distance-two vertices induce a matching together with isolated vertices: a vertex of that layer with two neighbours in the layer would have two common neighbours with $a$. Writing $m$ for the number of edges of the matching gives $n = 7+h$, $e = 8+h+m$ and $T = m$, so $S = 16+2h-m = 2n+2-m$. Thus $S \leq 2n + 2$, with equality if and only if $m = 0$. Equality means that $G$ is the graph $\Gamma_h$ formed by the 6-cycle, the vertex $a$ on one antipodal pair, and $h$ leaves at $a$.

The graph $\Gamma_h$ lies in the class, for every $h \geq 0$. A leaf has degree one. Every other cycle through an outside vertex uses both edges from $a$ to the cycle, and each of the two $a$-paths along the cycle has length three, so every such cycle has length five. There is no 4-cycle. There is no induced $P_6$. A leaf, having degree one, occupies an end of any path that contains it, and that path meets $a$ next. Two leaves therefore determine a path on only three vertices. An induced path that contains one leaf and five further vertices would restrict to an induced path of order five in $\Gamma_0$ that starts at $a$. No such path exists: the two neighbours of $a$ on the cycle are an antipodal pair, the path $a, c_i, c_{i+1}, c_{i+2}, c_{i+3}$ has the chord $ac_{i+3}$, and every shorter continuation from $a$ through one neighbour stops at the vertex before the antipode, because the next cycle vertex is adjacent to $a$. An induced path of $\Gamma_0$ that avoids $a$ lies on the 6-cycle and has order at most five. An induced path of $\Gamma_0$ that contains $a$ in its interior has $c_i$ and $c_{i+3}$ as the two neighbours of $a$ on the path. Each side can then take at most one further cycle vertex: the next vertex on either arc is adjacent to the antipode already used. The resulting path has order at most five. Thus $\Gamma_h$ is strong $C_4$-free and $P_6$-free, and it attains $S = 2n + 2$.

If $k \in \{2, 3\}$, then $h \leq 1$: two vertices of the second layer would both be adjacent to two vertices of $A$. There are no triangles. The second-layer vertex, if it exists, is adjacent to every vertex of $A$ and to nothing on the cycle, $A$ is independent, and each vertex of $A$ contributes two edges to the cycle. Thus $n = 6 + k + h$ and $S = 2(6 + 2k + kh)$. The four possibilities are

| $(k, h)$ | $(2, 0)$ | $(2, 1)$ | $(3, 0)$ | $(3, 1)$ |
|---|---|---|---|---|
| $S/n$ | $5/2$ | $8/3$ | $8/3$ | $3$ |

The four pairs are the graphs of Figure 1, with $(3, 0)$ and $(2, 1)$ both equal to $P - z$. Write $a_i$ for the vertex adjacent to the antipodal pair $\{c_i, c_{i+3}\}$. The pair $(k, h) = (3, 1)$ is the graph in panel (c): the three vertices $a_0, a_1, a_2$ and a vertex $z$ adjacent to all three. It is cubic of order 10. It is triangle-free, and the trace rules give no 4-cycle, so its girth is at least 5. Its diameter is 2. Indeed every pair of cycle vertices is at distance at most 2 along the cycle or through the corresponding $a_i$; the vertex $z$ meets every $a_i$; and a non-neighbour of $a_i$ on the cycle is adjacent to $c_i$ or to $c_{i+3}$. Suppose $v_0 \cdots v_5$ were an

induced path. Diameter 2 gives $v_0$ a common neighbour with each of $v_3, v_4, v_5$. Those neighbours lie off the path, because the only neighbour of $v_0$ on the path is $v_1$ and the path is induced. They are distinct: a vertex adjacent to two of $v_3, v_4, v_5$ would form a triangle with a path edge, or a 4-cycle with the length-2 subpath $v_3v_4v_5$. This is impossible: $v_0$ has degree 3, so it has only two neighbours off the path. Thus the graph is $P_6$-free. It is the Petersen graph $P$, and $\ell(P) = 3$.
Deleting $z$ leaves $(k, h) = (3, 0)$, the graph $P-z$ in panel (b). The pair $(2, 1)$ is isomorphic to the same graph. In the labelling with $A = \{a_0, a_1\}$ and second-layer vertex $z$, the cycle $c_0, a_0, c_3, c_4, a_1, c_1$ has its three antipodal pairs served by $c_5$, $z$ and $c_2$, and no vertex remains. Both graphs therefore have $\ell = 8/3$, and both are isomorphic to $P - v$. Induced subgraphs of $P$ are strong $C_4$-free and $P_6$-free. The pair $(2, 0)$ is the graph $H_8$ in panel (a), with $\ell = 5/2$. The three choices of the unused antipodal pair differ by a rotation, so this graph is unique up to isomorphism. Its order is 8, so it is neither $P$ nor $P - v$. For $k \in \{2, 3\}$ one has $n = 6 + k + h \leq 10$.
*Case 2: an induced five-cycle and no induced six-cycle.* Write $C = c_0 \cdots c_4c_0$. Every vertex at distance one from $C$ has a singleton or consecutive-pair trace: any other pair of contacts lies at distance two on $C$ and forms a 4-cycle. If a vertex $x$ were at distance two, with a neighbour $a$ in the first layer, the path $x, a$ followed by four consecutive cycle vertices, beginning at one contact of $a$ and omitting the second contact if $a$ has one, would be an induced $P_6$. Thus $C$ dominates $G$. Write $h$ for the number of vertices with singleton trace and $j$ for the number with a consecutive-pair trace. At most one vertex carries each of the five pair traces, since two would have two common neighbours, so $j \leq 5$. The edges outside $C$ are of the following kinds.

- Two singleton vertices with different supports are nonadjacent. Neighbouring supports give a 4-cycle. Supports at distance two give an induced 6-cycle along the longer arc of $C$. Edges within one singleton class form a matching, because two neighbours in that class would be two common neighbours of the supporting cycle vertex. Let $m$ be the number of these edges.
- No two pair vertices are adjacent. From their two distinct support edges one can choose adjacent cycle vertices, one from each edge, and those two vertices together with the pair vertices form a 4-cycle.
- A singleton supported at $c_i$ meets only a pair vertex supported on the opposite edge $\{c_{i+2}, c_{i+3}\}$. Every other pair edge contains a neighbour of $c_i$ and yields a 4-cycle. That pair vertex meets at most one vertex of the singleton class, again by the common-neighbour bound. Let $b$ be the number of singleton-to-pair edges, so $b \leq \min(h, j)$.

Consequently $n = 5+h+j$ and $e = 5+h+2j+m+b$. The triangles are the $j$ pair triangles and the $m$ matching triangles, so $T = j + m$ and

$$S = 2e - 3T = 2n + 2b - j - m.$$

The integer $b$ is at most 1. Suppose two singleton–pair edges are present. They use two distinct singleton supports. The opposite edge of $c_i$ is $\{c_{i+2}, c_{i+3}\}$. As $i$ varies, these five edges exhaust $C$, and each pair class contains at most one vertex. The graph is $P_6$-free and strong $C_4$-free, and the present case assumes that it has no induced $C_6$, so Lemma 2.9 applies. Under that lemma the set $W_i$ is the singleton class at $c_i$, and the set $X_{i,i+1}$ is the consecutive-pair class on the edge $\{c_i, c_{i+1}\}$. If the two singleton supports lie at distance 2 on $C$, they are $W_i$ and $W_{i+2}$ for some $i$, and both are nonempty. Lemma 2.9 states that at least one of those two sets is empty. If the supports lie at distance 1, take them to be $c_i$ and $c_{i+1}$. The opposite edge of $c_i$ is $\{c_{i+2}, c_{i+3}\}$ and the opposite edge of $c_{i+1}$ is $\{c_{i+3}, c_{i+4}\}$. Both pair classes are nonempty, one for each of the two edges, and they are $X_{i+2,i+3}$ and $X_{i+3,i+4}$. Lemma 2.9 states that at least one of $X_{j,j+1}$ and $X_{j+1,j+2}$ is empty for every $j$; the index $j = i + 2$ is this pair of classes. Either alternative contradicts the existence of both edges, so $b \leq 1$. The third emptiness assertion of Lemma 2.9 is not required for the count.
If $b = 0$, then $S = 2n - j - m \leq 2n$. If $b = 1$, then $j \geq 1$ because $b \leq \min(h, j)$, so $S = 2n + 2 - j - m \leq 2n + 1$. In this case $S \leq 2n + 1$.
*Case 3: no holes.* The graph is chordal, and every nonempty induced subgraph has a simplicial vertex. The neighbourhood of a simplicial vertex is a clique. The degree is at most two: four vertices of a clique contain a 4-cycle. A simplicial vertex of a connected chordal graph on at least two vertices is not a cut vertex, and the graph left by deleting it is chordal and strong $C_4$-free. In the graph present at the deletion, the loss is $L(v) = 2d(v) - 3t(v)$. Degree one gives loss 2. Degree two, with the two neighbours adjacent, gives one triangle and loss 1. Deleting vertices until a single vertex remains, one has $S \leq 2(n - 1) = 2n - 2$.
The three cases give the values of $M(n)$. A graph of order $n \geq 11$ cannot realise $k \in \{2, 3\}$ in Case 1, and $k = 0$ forces $G = C_6$. The only remaining graphs in Case 1 are the graphs $\Gamma_h$ and the graphs with the same first layer and $m \geq 1$. The latter have $S \leq 2n + 1$. Cases 2 and 3 give $S \leq 2n + 1$ and $S \leq 2n - 2$. Thus $M(n) = 2 + 2/n$, attained only by $\Gamma_{n-7}$.
At order 10 the only graph in the list with $S > 2 \cdot 10 + 2$ is $P$, with $S = 30$. At order 9 the only such graph is $P - v$, with $S = 24$, whether it arises as $(k, h) = (3, 0)$ or as $(k, h) = (2, 1)$. At order 8 the only such graph is $H_8$, with $S = 20$. At order 7

one has $h = 0$ in the equality case of $k = 1$, so the unique graph with $S = 16$ is $\Gamma_0$, while Cases 2 and 3 give at most 15 and 12. Since $8/3 < 3$ and $2 + 2/n < 3$ for $n \geq 11$, the bound $S \leq 3n$ holds for every order $n \geq 2$, with equality only for $P$. The case division does not use $n \geq 7$. At order 4 there is no cycle of length five or six, so only the chordal bound applies and $S \leq 6$. Equality holds for a tree, and the only trees on four vertices are $P_4$ and $K_{1,3}$. At order 6 the induced 6-cycle branch contributes only $C_6$, with $S = 12$. In the induced 5-cycle branch one has $h + j = 1$, hence $b = 0$ and $S = 12 - j - m \leq 12$, with equality precisely when $j = m = 0$ and $h = 1$, the graph $C_5$ with one leaf. The chordal branch contributes at most 10. Thus the maximum is 12, attained by exactly two graphs. □

*Remark 3.7.* Karthick and Maffray describe the structure of graphs with no induced $P_6$ and no induced 4-cycle, and deduce the colouring bound $\chi \leq \lceil 5\omega/4 \rceil$ [13]. Their Theorems 3.2 and 3.3 describe the neighbourhood, on an induced $C_5$ or $C_6$, of a vertex outside the cycle. Specialising the 6-cycle description to graphs with no 4-cycle subgraph excludes a consecutive triple, a consecutive quadruple and the full cycle as a support, and leaves an antipodal pair. That is the trace statement at the beginning of Case 1. Part (d) of their Theorem 3.2 is recorded in full as Lemma 2.9, and Case 2 invokes its first two emptiness assertions to bound the number of singleton–pair edges. The constraint on the second layer, the bound $h \leq 1$ when $k \in \{2, 3\}$, and the exact function $M(n)$ are further arguments. They are not consequences of the local support list, and the colouring theorem is not a substitute for the extremal statement under the subgraph prohibition. Chen's Remark 7.24, cited in the introduction, likewise stops at five frames of orders 6 through 9 inside RES $\wedge$ $(C_6)$. It does not give the second-layer classification or $M(n)$.

*Remark 3.8.* Brandstädt and Hoàng analysed $(P_6, C_4)$-free graphs through clique cutsets [1]. In that language $C_4$ is an induced cycle. A graph in the class with no clique cutset has every induced $C_5$ dominating, and the presence of an induced $C_6$ that fails to dominate forces the graph to be the join of a blow-up of the Petersen graph and a clique, possibly empty. Theorem 3.6 is a different statement. The prohibition here is that of a 4-cycle as a subgraph, and the conclusion is the exact maximum of $S$ over every connected graph in the resulting class, including graphs that have clique cutsets. A nontrivial blow-up of the Petersen graph, and a join of the Petersen graph with a nonempty clique, each contain a 4-cycle subgraph, so neither construction enters the class. Chen's Remark 7.24 is a restriction of the same kind: the five frames do not include the Petersen graph as the extremal graph of order 10, and they do not include $\Gamma_{n-7}$.

*Remark 3.9.* An alternative proof of $S \leq 3n$, with equality only for the Petersen graph, fixes a vertex of eccentricity at most 2 by Lemma 3.5 and charges the cross edges in the second distance layer. The case division above already gives that inequality and the equality case, so the charging argument is not repeated.

## 4 | Radius four

**Lemma 4.1.** *Let $G$ be connected, strong $C_4$-free and $P_8$-free. If an induced 8-cycle dominates $G$ and $r(G) \geq 4$, then $G = C_8$.*

*Proof.* Write $C = c_0c_1 \cdots c_7c_0$, with indices modulo 8. For an outside vertex $y$ the trace $A_y = N(y) \cap V(C)$ is nonempty. The trace $A_y$ is not contained in any three consecutive vertices of $C$. Suppose $A_y \subseteq \{c_i, c_{i+1}, c_{i+2}\}$. The ends $c_i$ and $c_{i+2}$ are not both in $A_y$, since $y, c_i, c_{i+1}, c_{i+2}$ would be a 4-cycle. Thus $A_y$ is a singleton or a consecutive pair. A singleton $\{c_i\}$ gives the induced path

$$y, c_i, c_{i+1}, c_{i+2}, c_{i+3}, c_{i+4}, c_{i+5}, c_{i+6}.$$

A consecutive pair $\{c_i, c_{i+1}\}$ gives the induced path

$$y, c_i, c_{i-1}, c_{i-2}, c_{i-3}, c_{i-4}, c_{i-5}, c_{i-6},$$

in which the omitted contact $c_{i+1}$ does not lie on the path. Either path is an induced $P_8$.
Fix a cycle vertex, say $c_0$. The vertices at cycle distance greater than two from $c_0$ are the three consecutive vertices $c_3, c_4, c_5$. The trace of any outside vertex meets the complementary set $\{c_7, c_0, c_1, c_2, c_6\}$, so every outside vertex is within distance three of $c_0$. The same holds with $c_0$ replaced by any vertex of $C$.
Suppose $w$ lies outside $C$, and write $A = N(w) \cap V(C)$. No two vertices of $A$ are at cycle distance two, for they would form a 4-cycle with $w$. A set of vertices of $C$ in which every two distinct vertices are consecutive has size at most two, and every set of size at most two that is a singleton or a consecutive pair is contained in a triple of consecutive vertices. Both possibilities have been excluded. Therefore $A$ contains two vertices at cycle distance three or four. Rotating, take them to be $c_0$ and $c_d$ with $d \in \{3, 4\}$.
The antipode of $c_0$ is $c_4$. If $d = 4$, the path $c_0wc_4$ has length two. If $d = 3$, then $c_3 \sim c_4$, so $c_0wc_3c_4$ has length three. Every other vertex of $C$ is already at cycle distance at most three from $c_0$. Every outside vertex is within distance three of $c_0$. Thus the eccentricity of $c_0$ is at most three, contradicting $r(G) \geq 4$.

Hence $C$ has no vertex outside it, and $G = C_8$. □

*Remark 4.2.* The proof of Lemma 4.1 does not use a machine enumeration. An independent count, with the labelled cycle fixed, is the script `paper/scripts/c8_configurations.py`. A trace is admissible when the graph formed by the cycle and one vertex of that trace has no 4-cycle subgraph and no induced $P_8$. Configurations are built by adding admissible traces in increasing bitmask order, joining the new vertex to an arbitrary subset of the outside vertices already chosen, and discarding any graph that acquires a 4-cycle subgraph or an induced $P_8$. The script finds 32 admissible nonempty traces and 790 configurations, including the bare cycle. By order 8, ... , 13 the counts are 1, 32, 262, 384, 107, 4. It finds 73 isomorphism classes, and only the bare cycle has radius at least four. That count is not a hypothesis of Theorem 4.3.

**Theorem 4.3.** *A connected graph $G$ is strong $C_4$-free and satisfies $r(G) = 4$ and $p(G) = 7$ if and only if either $G \cong C_8$, or $G$ is obtained from a 6-cycle $C = v_0v_1 \cdots v_5$ by the following construction and by no further vertices or edges. For each $i$ there is a nonempty set $C_i$ of vertices outside $C$. Every vertex of $C_i$ is adjacent to $v_i$ and to no other vertex of $C$. There is no edge between $C_i$ and $C_j$ for $i \neq j$, and $G[C_i]$ is a matching together with isolated vertices. For each edge of $C$ there is at most one further vertex, adjacent precisely to the two ends of that edge.*

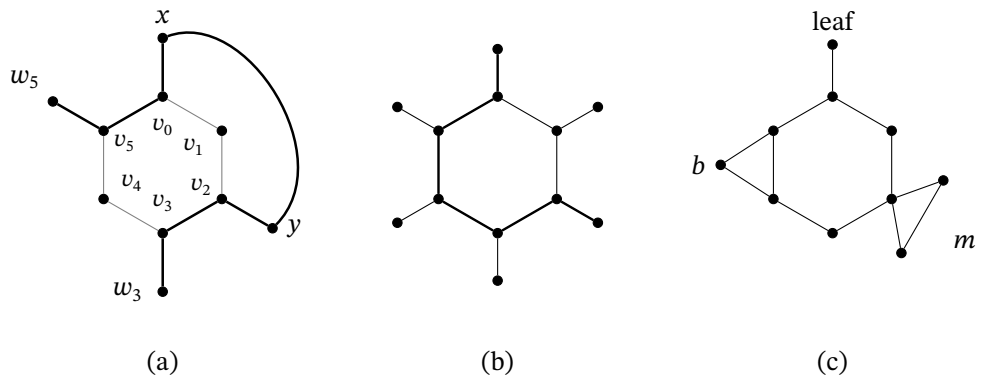



**FIGURE 3** | The radius-four case of Theorem 4.3. (a) The heavy path is $w_5, v_5, v_0, x, y, v_2, v_3, w_3$. The vertices $x$ and $y$ have singleton traces $\{v_0\}$ and $\{v_2\}$, and $w_5$ and $w_3$ are the distance-four witnesses. The edge $xy$ curves outside the cycle, clear of $v_1$ and of every other cycle vertex. The path is induced because the 6-cycle is induced, each off-cycle vertex meets only its designated cycle vertex, and the neighbourhoods of the two witnesses are constrained. (b) One leaf at each cycle vertex gives $S = 2n$. The heavy path has seven vertices. (c) On the same cycle, only the local contributions to $b$ and $m$ are drawn; the other attachments are omitted. The figure shows an isolated leaf, a matching edge inside one set $C_i$, and a vertex glued to a cycle edge. The leaf does not contribute to $b + m$. The matching edge and the glued vertex each increase $2n - S = b + m$ by 1.

**Lemma 4.4.** *Let $G$ be obtained from a 6-cycle $C$ by the construction of Theorem 4.3. Let $P$ be an induced path in $G$ of order at least three. Every vertex of $P$ that does not lie on $C$ is an endpoint of $P$. In particular $P$ contains at most two vertices outside $C$, and any such vertex is an endpoint of $P$.*

*Proof.* Let $z$ be a vertex of $P$ outside $C$. If $z$ is an isolated vertex of $G[C_i]$ for some $i$, then $z$ has degree one, so $z$ is an endpoint of every path that contains it.
If $z$ lies on a matching edge of $G[C_i]$, the neighbourhood of $z$ is $\{v_i, z'\}$, where $z'$ is the other end of that edge, and this neighbourhood is a clique. Suppose $z$ were internal on $P$. An internal vertex has two neighbours on $P$. Those neighbours are then $v_i$ and $z'$, which are adjacent, and the path uses the route $v_izz'$ or $z'zv_i$. The edge $v_iz'$ joins the two neighbours of $z$ and skips $z$, so it is a chord of $P$. An induced path has no chord. Thus $z$ is an endpoint of $P$. The same chord shows that $P$ does not contain $z$, $z'$ and $v_i$ together.
If $z$ was added on a cycle edge, the neighbourhood of $z$ is that edge, again a clique. The same chord argument shows that $z$ is an endpoint of $P$, and that $P$ does not contain both ends of the cycle edge together with $z$.
A path has two endpoints. Every vertex of $P$ outside $C$ is one of them, so there are at most two such vertices. □

*Proof of Theorem 4.3.* First suppose that $G$ is strong $C_4$-free, with $r(G) = 4$ and $p(G) = 7$. Theorem 2.4 gives $p(G) \geq 7$, which agrees with the hypothesis. Apply Theorem 2.5 with $k = 8$. The hypotheses hold: $k \geq 4$, and $G$ is connected and has no induced path on eight vertices because $p(G) = 7$. The conclusion is a connected dominating set $X$ such that $G[X]$ is $P_6$-free or $G[X] \cong C_8$.
If $G[X] \cong C_8$, Lemma 4.1 gives $G = C_8$.

If $G[X]$ is $P_6$-free, apply Lemma 3.5 inside $G[X]$. A vertex of eccentricity at most two in $G[X]$ has eccentricity at most three in $G$, because $X$ dominates $G$, contradicting $r(G) = 4$. Therefore $G[X] = C_6$. Let $C = v_0 \cdots v_5$ be this dominating induced 6-cycle, indices modulo 6, and let $W = V(G) \setminus V(C)$.

*Traces.* As in the proof of Lemma 3.4, strong $C_4$-freeness allows only three kinds of trace $N(w) \cap C$: a single vertex, a consecutive pair $\{v_i, v_{i+1}\}$, or an antipodal pair. If some $w$ has trace $\{v_i, v_{i+3}\}$, then $v_i$ is within distance two of every vertex of $C$, hence within distance three of every vertex of $G$, a contradiction. So there are no antipodal traces. Let $C_i$ be the set of vertices with trace $\{v_i\}$.

*Witnesses.* Fix $i$. Since $r(G) = 4$, some vertex is at distance at least four from $v_{i+3}$. Every vertex of $C$ is within distance three of $v_{i+3}$, and a vertex $w \in W$ is at distance at most $1 + \min\{d_C(v_{i+3}, x) : x \in N(w) \cap C\}$ from $v_{i+3}$, which is at most three unless the trace of $w$ is $\{v_i\}$. So there is $w_i \in C_i$ at distance four from $v_{i+3}$. Every neighbour $z \in W$ of $w_i$ lies in $C_i$: if $z$ had a neighbour among $v_{i+2}, v_{i+3}, v_{i+4}$, then $d(w_i, v_{i+3}) \le 3$; if $z$ had the neighbour $v_{i-1}$ or $v_{i+1}$, then $w_i, v_i, v_{i\pm1}, z$ would be a 4-cycle. So $N(w_i) \subseteq \{v_i\} \cup C_i$.

*Edges in $W$.* Figure 3 records the two configurations used below. Let $x, y \in W$ be adjacent.

- $x \in C_0$, $y \in C_{\pm1}$: $x, v_0, v_{\pm1}, y$ is a 4-cycle.
- $x \in C_0$, $y \in C_2$: the path $w_5, v_5, v_0, x, y, v_2, v_3, w_3$ of Figure 3(a) is an induced $P_8$. The 6-cycle is induced, so it contributes no chord among $v_5, v_0, v_2, v_3$. Each off-cycle vertex meets the cycle only at its designated vertex: $w_5$ at $v_5$, $x$ at $v_0$, $y$ at $v_2$ and $w_3$ at $v_3$. The witness neighbourhoods, already proved, satisfy $N(w_5) \subseteq \{v_5\} \cup C_5$ and $N(w_3) \subseteq \{v_3\} \cup C_3$, so neither witness is adjacent to $x$, to $y$, or to the other witness. These three facts leave no chord.
- $x \in C_0$, $y \in C_3$: $w_1, v_1, v_0, x, y, v_3, v_4, w_4$ is an induced $P_8$. By symmetry, vertices of $C_i$ and $C_j$ with $i \ne j$ are never adjacent.
- $x$ with trace $\{v_0, v_1\}$ and $y \in C_j$: for $j \in \{0, 1\}$ the vertices $v_0, v_1, x, y$ span a 4-cycle; for $j = 5$ (respectively $j = 2$) the cycle $x, v_0, v_5, y$ (respectively $x, v_1, v_2, y$) is a 4-cycle; for $j = 3$, $w_5, v_5, v_0, x, y, v_3, v_2, w_2$ is an induced $P_8$, and $j = 4$ is symmetric.
- $x, y$ with consecutive-pair traces $\{v_0, v_1\}$ and $\{v_j, v_{j+1}\}$: $j = 0$ gives two common neighbours; $j = \pm1$ and $j = \pm2$ give 4-cycles as above; $j = 3$ gives the induced $P_8$ $w_2, v_2, v_1, x, y, v_4, v_5, w_5$.

The remaining paths displayed above are induced for the same three reasons, applied to the witnesses and traces named in those bullets. So every edge inside $W$ joins two vertices of the same class $C_i$, and these edges form a matching. A consecutive-pair vertex is adjacent to no vertex of any $C_j$ and to no other consecutive-pair vertex. Two vertices with the same consecutive-pair trace would have two common neighbours, so each cycle edge carries at most one such vertex, and that vertex is adjacent precisely to the two ends of the edge. Each class $C_i$ is nonempty. This is the stated construction.

Conversely, let $G$ be $C_8$, or let $G$ be built from $C = v_0 \cdots v_5$ as in the statement. The cycle $C_8$ has radius 4, induced-path order 7 and no 4-cycle subgraph. Now consider the construction on $C$. There is no edge between distinct sets $C_i$, and every vertex of $C_i$ meets $C$ only at $v_i$. A cycle that meets $C_i$ is therefore contained in $G[C_i \cup \{v_i\}]$. In that subgraph the edges are a star at $v_i$ together with a matching, so the only cycles are the triangles formed by $v_i$ and one matching edge.

Let $y$ be a vertex added on the edge $\{v_j, v_{j+1}\}$, so $N(y) = \{v_j, v_{j+1}\}$. Every cycle through $y$ uses both edges at $y$, and the rest of the cycle is a $v_j$–$v_{j+1}$ path avoiding $y$. A 4-cycle through $y$ is then a path of length two between $v_j$ and $v_{j+1}$ that avoids $y$: the two ends of the edge have a common neighbour other than $y$. No vertex of any $C_i$ is adjacent to two vertices of $C$, and the construction places at most one vertex on each cycle edge. Those two ends therefore have no common neighbour other than the vertex added on that edge. Hence no 4-cycle passes through $y$. The only cycle that uses no vertex outside $C$ is $C$ itself. Thus $G$ has no 4-cycle subgraph.

Let $P$ be an induced path of order at least three. By Lemma 4.4, every vertex of $P$ outside $C$ is an endpoint of $P$, so every other vertex of $P$ lies on $C$. The cycle $C$ has no induced path on six vertices. Hence $p(G) \le 7$. Since each $C_i$ is nonempty, one may choose $x \in C_0$ and $y \in C_2$. The path $x, v_0, v_5, v_4, v_3, v_2, y$ is induced: $x$ and $y$ meet the path only at $v_0$ and $v_2$, and $C$ contributes no chord. Thus $p(G) = 7$.

There is no edge between distinct sets $C_i$, and a vertex added on a cycle edge has degree two on that edge, so it does not shorten distances along $C$. The distance from $v_i$ to $v_{i+3}$ is therefore three. A vertex of $C_{i+3}$ is at distance four from $v_i$, and the eccentricity of $v_i$ is at least four. It is at most four: every vertex of $C_j$ is at distance $d_C(v_i, v_j) + 1 \le 4$ from $v_i$, and every vertex added on a cycle edge is within distance four of $v_i$. A vertex of $C_i$ is at distance five from a vertex of $C_{i+3}$. A vertex $y$ added on the edge $\{v_j, v_{j+1}\}$ has only those two neighbours. The distances from $v_j$ and from $v_{j+1}$ to $v_{j+3}$ are 3 and 2, neither neighbour of $y$ is adjacent to $v_{j+3}$, and $y, v_{j+1}, v_{j+2}, v_{j+3}$ is a path of length three, so a vertex of $C_{j+3}$ is at distance four from $y$. Every eccentricity is at least four, and $r(G) = 4$. □

**Corollary 4.5.** *Let $G$ be connected and strong $C_4$-free with $r(G) = 4$ and $p(G) = 7$. In the construction of Theorem 4.3, let $b$ be the number of vertices added on edges of the 6-cycle and let $m$ be the number of matching edges in the sets $C_i$. Then $S(G) = 2n - b - m$. In particular $S(G) \le 2n$, so $\ell(G) \le 2$, and* RAD(4) *holds.*

*Proof.* For $G \cong C_8$ one has $n = 8, e = 8, T = 0$ and $S = 16 = 2n$, which is the formula with $b = m = 0$. In the construction, write $a = \sum_i |C_i|$. Then $n = 6 + a + b$. The edges are the six cycle edges, $a$ edges from the sets $C_i$ to the cycle, $m$ matching edges and $2b$ edges incident with the vertices added on cycle edges, so $e = 6 + a + 2b + m$. Each matching edge forms one triangle with its cycle vertex, and each added vertex forms one triangle with its cycle edge, so $T = b + m$. Lemma 2.1(3) gives $S = 2e - 3T = 2n - b - m$. □

**Corollary 4.6.** *Let $G$ be connected and strong $C_4$-free with $r(G) = 4$ and $p(G) = 7$. Then $S(G) = 2n$ if and only if $G \cong C_8$, or $G$ is obtained from $C_6$ by attaching at least one leaf to each cycle vertex.*

*Proof.* By Corollary 4.5, $S = 2n$ if and only if $G \cong C_8$ or, in the construction, $b = m = 0$. In the latter case every vertex outside the cycle is a leaf, and each $C_i$ is nonempty, so each cycle vertex has at least one leaf. The converse is the case $b = m = 0$ of the construction, together with $C_8$, both of which have $S = 2n$ and satisfy the numerical hypotheses by Theorem 4.3. □

*Remark 4.7.* If $G$ is connected and strong $C_4$-free with $r(G) = 4$, $p(G) \ge 8$ and $\ell(G) \le 4$, then $p(G) \ge 8 = r(G) + 4 \ge r(G) + \ell(G)$. This comparison is not a structural description of $G$.

## 5 | Graphs of induced-path order at most six

In a triangle-free graph, a 4-cycle subgraph is an induced 4-cycle, and every clique has order at most two. The next classification is Lemma 3.4 of Chen and Xu [5]. Their family $\mathfrak{F} = \{F_0, F_0', F_1, \dots, F_{12}\}$ is defined by the constructions in the proofs of their Lemmas 3.1–3.3; $F_5$ is the Petersen graph, and the parameters $t$ and $t'$ attached to $F_2$ and $F_0'$ are positive integers. The edge counts below use those constructions rather than the accompanying figure.

**Lemma 5.1** (Chen and Xu [5, Lemma 3.4])**.** *Let $G$ be a connected $(P_7, C_4, C_3)$-free graph with no clique cutset. If $G$ is not complete, then $G \in \mathfrak{F}$.*

The members of $\mathfrak{F}$ admit the following descriptions, taken from those proofs.

- $F_1$ is $C_5$. The graph $F_2$ is obtained from a 5-cycle $v_1 \cdots v_5$ by adding stable sets $A_1$ and $A_3$ of equal positive order $t$, with a perfect matching between them, each vertex of $A_1$ adjacent only to $v_1$ and its match, and each vertex of $A_3$ adjacent only to $v_3$ and its match.
- On the same 5-cycle, $F_3$ has three further vertices inducing the path $a_3a_1a_4$, with $a_i$ adjacent to $v_i$. The graph $F_4$ has four further vertices inducing the path $a_5a_3a_1a_4$, with the same spokes. The graph $F_5$ is the Petersen graph.
- $F_0$ is $C_6$. The graph $F_0'$ is obtained from a 6-cycle $v_1 \cdots v_6$ by adding stable sets $Y_1$ and $Y_4$ of equal positive order $t'$, with a perfect matching between them, each vertex of $Y_1$ adjacent only to $v_1$ and its match, and each vertex of $Y_4$ adjacent only to $v_4$ and its match.
- $F_6$ is $C_7$. On a 7-cycle $v_1 \cdots v_7$, write $x_i$ for a vertex with neighbourhood $\{v_i, v_{i+3}\}$ on the cycle. Then $F_7$ is the cycle plus $x_1$; $F_8$ is the cycle plus nonadjacent $x_1, x_2$; $F_9$ is $F_8$ plus a vertex adjacent only to $x_1$ and $x_2$; $F_{10}$ is the cycle plus nonadjacent $x_1, x_5$; $F_{11}$ is the cycle plus a stable set $\{x_1, x_2, x_5\}$; and $F_{12}$ is $F_{11}$ plus a vertex adjacent only to $x_1$ and $x_2$.

Chen and Xu record that $F_0, F_1$, the graph $F_2$ with $t = 1$, $F_3$ and $F_4$ are induced subgraphs of the Petersen graph, and that $F_6, \dots, F_{11}$ are induced subgraphs of $F_{12}$. The graphs $F_6, \dots, F_{12}$ are the extensions of $C_7$ by vertices of cycle-distance three. The graphs $F_2$ and $F_0'$ are the matching extensions of one pair of opposite vertices on $C_5$ and on $C_6$.

**Lemma 5.2.** *Every graph in $\mathfrak{F}$ other than the Petersen graph satisfies $2e \le 3n - 3$. The Petersen graph satisfies $2e = 3n$.*

*Proof.* The constructions give the following orders and sizes. For $F_2$ and $F_0'$ the parameters $t, t'$ are at least one.

| Graph | $n$ | $e$ | $2e$ | $3n-2e$ |
|---|---|---|---|---|
| $F_1 = C_5$ | 5 | 5 | 10 | 5 |
| $F_2$ | $5+2t$ | $5+3t$ | $10+6t$ | 5 |
| $F_3$ | 8 | 10 | 20 | 4 |
| $F_4$ | 9 | 12 | 24 | 3 |
| $F_5$ | 10 | 15 | 30 | 0 |
| $F_0 = C_6$ | 6 | 6 | 12 | 6 |
| $F_0'$ | $6+2t'$ | $6+3t'$ | $12+6t'$ | 6 |
| $F_6 = C_7$ | 7 | 7 | 14 | 7 |
| $F_7$ | 8 | 9 | 18 | 6 |
| $F_8, F_{10}$ | 9 | 11 | 22 | 5 |
| $F_9, F_{11}$ | 10 | 13 | 26 | 4 |
| $F_{12}$ | 11 | 15 | 30 | 3 |

The five edges of $F_3$ beyond $C_5$ are the three spokes and the two edges of $a_3a_1a_4$. The seven edges of $F_4$ beyond $C_5$ are the four spokes and the three edges of $a_5a_3a_1a_4$. Each vertex $x_i$ contributes two edges, and each vertex placed on a pair $x_ix_j$ contributes two further edges. The Petersen graph is the only row with difference zero. □

**Lemma 5.3.** *Let $F$ be connected and strong $C_4$-free, and let $s$ be a vertex at which no induced $P_4$ starts. Write $m = e(F[N(s)])$. Then $S(F) \le 2(|F|-1) - m$.*

*Proof.* A vertex at distance 3 from $s$ would end a geodesic, hence an induced $P_4$, starting at $s$. So $V = \{s\} \cup N_1 \cup N_2$ with $N_i$ the $i$-th distance layer. Each $y \in N_2$ has a unique parent $a \in N_1$. A neighbour $z \in N_2$ of $y$ with a different parent would give the induced path $sayz$. So $y$ has at most one neighbour in $N_2$, a sibling (two siblings would be common neighbours of $a$ and $y$), and $N(y)$ is a clique; thus $\lambda(y) = 1$. For $a \in N_1$ we have $N(a) \subseteq \{s, \bar{a}\} \cup C_a$, where $C_a$ is the set of children of $a$ and $\bar{a}$ is the neighbour of $a$ in $N_1$, if any. Since $s$ and $\bar{a}$ lie in one unit, $\lambda(a) \le 1 + |C_a|$. Finally $\lambda(s) = |N_1| - m$. Summing, $S \le (|N_1| - m) + (|N_1| + |N_2|) + |N_2| = 2(|F|-1) - m$. □

**Lemma 5.4.** *Let $G$ be a connected triangle-free strong $C_4$-free $P_7$-free graph of order $n$. Then $G$ is obtained from a connected triangle-free strong $C_4$-free $P_7$-free graph $Q$ with no clique cutset by attaching trees at single vertices. The core $Q$ is $K_1$, $K_2$, or a graph in the list of Lemma 5.2. Writing $h = |V(Q)|$ and $\eta(Q) = 3h - 2e(Q)$, one has $e(G) = e(Q) + (n-h)$ and*

$$3n - 2e(G) = \eta(Q) + n - h.$$

*Deleting the vertices of one attached tree, other than its root on $Q$, deletes exactly as many edges as vertices.*

*Proof.* If $G$ itself has no cut vertex, Lemma 2.10 gives that $G$ has no clique cutset. The hypotheses of that lemma are the present hypotheses together with the absence of a cut vertex. Lemma 5.1 then applies. Its hypotheses are that $G$ is connected, that $G$ has no induced $P_7$, that $G$ has no induced $C_4$ and no triangle, and that $G$ has no clique cutset. Connectedness and the absence of a clique cutset hold by the case and by Lemma 2.10. The graph is $P_7$-free by hypothesis. A strong $C_4$-free graph has no induced $C_4$, and a triangle-free graph has no $C_3$. If $G$ is not complete, Lemma 5.1 places it in the family $\mathfrak{F}$, hence on the list of Lemma 5.2. The only complete triangle-free graphs are $K_1$ and $K_2$. In every subcase one may take $Q = G$. The identities are then immediate.

Suppose $s$ is a cut vertex. Path concatenation supplies a component $C$ of $G - s$ such that every induced path of $G[C \cup \{s\}]$ that starts at $s$ has order at most three. The first distance layer of this side is independent, since an edge there would form a triangle with $s$. There is no vertex at distance three or more from $s$, for such a vertex would end an induced $P_4$ at $s$. A vertex of the second layer has a unique parent: a second parent, together with $s$, would span a 4-cycle. It has no neighbour in its layer. An edge joining vertices of distinct parents is the induced path of order four recorded in the proof of Lemma 5.3, and an edge joining two children of the same parent forms a triangle with that parent. Thus $G[C \cup \{s\}]$ is a tree. Deleting the $k = |C|$ vertices of the side deletes exactly $k$ edges. The graph that remains is connected, triangle-free, strong $C_4$-free and $P_7$-free. Iterate. The remaining graph $Q$ has no cut vertex, and the first paragraph describes it. Reversing the deletions attaches trees at single vertices. Each deleted vertex accounted for one edge, so $e(G) = e(Q) + (n-h)$, and the displayed identity follows. □

**Theorem 5.5.** *Let $G$ be a connected triangle-free strong $C_4$-free $P_7$-free graph. Then $2e(G) \le 3|V(G)|$, with equality if and only if $G$ is the Petersen graph. In particular every connected triangle-free strong $C_4$-free graph of induced-path order at most 6 satisfies $\mathrm{GEN}_3$, and so does every connected bipartite strong $C_4$-free graph of induced-path order at most 6. A bipartite graph in the class has strict inequality.*

*Proof.* Lemma 5.4 gives a core $Q$ of order $h$ and the identity

$$3n - 2e(G) = \eta(Q) + n - h.$$

Lemma 5.2 gives $\eta(Q) \geq 0$ for every graph on its list, with equality only for the Petersen graph. The cores $K_1$ and $K_2$ have deficits 3 and 4. The right-hand side is therefore nonnegative, and it vanishes if and only if $\eta(Q) = 0$ and $n = h$. Thus equality holds if and only if $G$ is the Petersen graph.
A bipartite graph is triangle-free. The Petersen graph is not bipartite, so the inequality is strict for every connected bipartite strong $C_4$-free $P_7$-free graph. □

Lemma 5.1 supplies the cores, the connected graphs in the class with no clique cutset. The next theorem does not reclaim that list. It gives the exact value of $f(n)$ and the equality cases on every connected triangle-free strong $C_4$-free $P_7$-free graph of order $n \geq 10$, including the graphs obtained by attaching trees to those cores.

**Theorem 5.6.** *Let $f(n)$ be the maximum of $2e(G)$ over connected triangle-free strong $C_4$-free $P_7$-free graphs of order $n$. For every $n \geq 10$,*

$$f(n) = 3n - \min\{n - 10,\ 5 + \mathbf{1}_{\{n \text{ even}\}}\}.$$

*Write $\mathcal{P}(n)$ for the isomorphism classes of graphs obtained from the Petersen graph by placing $n - 10$ leaves on a single vertex or on the two endpoints of a single edge. For $10 \leq n \leq 14$ the extremal graphs are exactly the classes in $\mathcal{P}(n)$. For $n = 15$ they are the classes in $\mathcal{P}(15)$ together with $F_2$, four classes in all. For $n = 16$ they are the classes in $\mathcal{P}(16)$, the graph $F_0'$, and the two graphs obtained from the order-15 graph $F_2$ by adding a leaf at a hub or at the midpoint of its unique 2-edge branch, seven classes in all. For odd $n \geq 17$ the unique extremal graph is $F_2$. For even $n \geq 18$ there are exactly three: $F_0'$, and the two graphs obtained from the order-$(n-1)$ graph $F_2$ by adding a leaf at a hub or at the midpoint of its unique 2-edge branch.*

*Proof.* Write $\eta(H) = 3|V(H)| - 2e(H)$ for a triangle-free graph $H$.
*Reduction to one core.* Lemma 5.4 supplies a core $Q$ of order $h$ and the identity $3n - 2e(G) = \eta(Q) + n - h$.
*Upper bound.* Every fixed row of the table in Lemma 5.2 satisfies $\eta \geq h - 10$. Each such graph has order at most 10 except $F_{12}$, which has order 11 and $\eta = 3$, and $3 \geq 11 - 10$. The same inequality holds for $K_1$ and $K_2$, with deficits 3 and 4. For every such core, $\eta(Q) + n - h \geq n - 10$. The two parameterised rows are $F_2$, of odd order and deficit 5, and $F_0'$, of even order and deficit 6. If the core is $F_2$ and the final order is even, then at least one vertex has been attached, so the displayed deficit is at least 6. If the core is $F_2$ and the final order is odd, the deficit is at least 5. If the core is $F_0'$, the deficit is at least 6. In every case

$$3n - 2e(G) \geq \min\{n - 10,\ 5 + \mathbf{1}_{\{n \text{ even}\}}\}.$$

*Attainment.* The value $f(n)$ is attained by the graphs named in the statement; the next paragraph shows that no other graph attains it. A leaf added to a triangle-free strong $C_4$-free graph creates neither a triangle nor a cycle. Realise the Petersen graph on the 2-element subsets of $\{1, 2, 3, 4, 5\}$, with two subsets adjacent when they are disjoint. This is the graph $P$ of Theorem 3.6. The symmetric group on the ground set acts on the vertices. The action is vertex-transitive and edge-transitive, and it carries any two nonadjacent vertices to $\{1, 2\}$ and $\{1, 3\}$. The path

$$\{1, 2\}, \{3, 4\}, \{1, 5\}, \{2, 3\}, \{1, 4\}$$

is an induced $P_5$ starting at $\{1, 2\}$, and

$$\{1, 2\}, \{3, 4\}, \{1, 5\}, \{2, 4\}, \{1, 3\}$$

is an induced $P_5$ with ends $\{1, 2\}$ and $\{1, 3\}$. Thus every vertex of $P$ is an end of an induced $P_5$, and every pair of nonadjacent vertices is the pair of ends of an induced $P_5$.
Place $a$ leaves at one end of an edge of $P$ and $b$ leaves at the other, with $a + b = n - 10$ and $a \geq b \geq 0$. The case $b = 0$ is the placement at a single vertex. The graph is connected, triangle-free and strong $C_4$-free. It has $15 + (n - 10)$ edges, so its deficit is $n - 10$. An induced path uses a leaf only as an end. A path that uses two leaves at the same vertex has order three. A path that uses one leaf at each end of the chosen edge is the edge itself with the two leaves attached: the ends of that edge are adjacent, so no longer induced path of $P$ has those ends. A path that uses one leaf restricts to an induced path of $P$ and has order at most six, because $P$ is $P_6$-free. Thus the graph is $P_7$-free. Vertex-transitivity and edge-transitivity give one isomorphism class for each pair $(a, b)$. Distinct pairs give nonisomorphic graphs: the Petersen graph is cubic, so the degrees of the vertices adjacent to a leaf are $3 + a$ and $3 + b$ when $b \geq 1$, and the single value $3 + a$ when $b = 0$. These are the classes in $\mathcal{P}(n)$, and they attain deficit $n - 10$.
For the deficit $5 + \mathbf{1}_{\{n \text{ even}\}}$, use $F_2$ at odd orders and $F_0'$ at even orders. In the descriptions preceding Lemma 5.2, each of these graphs consists of internally disjoint three-edge paths between two hubs, together with one additional two-edge path in the case of $F_2$. Every cycle uses two of those paths and therefore has length at least five. An induced path that contains at most one hub meets that hub in at most two of the paths and so has at most five vertices. An induced path that contains both hubs uses at most one path between them, a segment of at most three edges, and each end extends by at most one further edge before the next vertex meets the opposite hub and creates a chord. Its order is at most six. The table of Lemma 5.2 gives deficits 5 and 6 respectively. The smaller of the two deficits is therefore attained.

*Equality.* Write $\delta(n)$ for the minimum in the statement, and let $G$ satisfy $3n - 2e(G) = \delta(n)$. The identity $3n - 2e(G) = \eta(Q) + n - h$ and the upper bound force the core $Q$ to be the Petersen graph, $F_2$ or $F_0'$. Every other core in the table of Lemma 5.2, and each of $K_1$ and $K_2$, has $\eta(Q) - h \geq -8$, hence deficit at least $n - 8$. For $n \geq 10$ one has $n - 8 > \delta(n)$, since $\delta(n) \leq n - 10$.

If $Q$ is the Petersen graph, then $\eta(Q) = 0$, $h = 10$ and the deficit is $n - 10$. This equals $\delta(n)$ if and only if $10 \leq n \leq 16$. The vertices outside $Q$ form trees attached at single vertices of $Q$. A tree of depth at least two contains a path $y, a, s$ of length two ending at its attachment vertex $s$. Concatenating that path with an induced $P_5$ of $Q$ that starts at $s$ produces an induced $P_7$: the only edge from $\{y, a\}$ into $Q$ is $as$, and $a$ is not adjacent to the neighbour of $s$ on the $P_5$ because that neighbour and $a$ would form a triangle with $s$. Thus every attached tree is a star of leaves at its attachment vertex.

The set $A$ of vertices of $Q$ that are adjacent to at least one leaf is a clique. Suppose $u, v \in A$ are distinct and nonadjacent, and let $\ell_u$ and $\ell_v$ be leaves at $u$ and at $v$. The paragraph on attainment supplies an induced $P_5$ of $Q$ with ends $u$ and $v$; write it $u = q_0, q_1, q_2, q_3, q_4 = v$. The path $\ell_u, q_0, q_1, q_2, q_3, q_4, \ell_v$ has seven vertices. It is induced. The leaf $\ell_u$ is adjacent only to $q_0$, hence to none of $q_1, q_2, q_3, q_4, \ell_v$. The leaf $\ell_v$ is adjacent only to $q_4$, hence to none of $\ell_u, q_0, q_1, q_2, q_3$. The path $q_0q_1q_2q_3q_4$ is induced, so it contributes no chord. This is an induced $P_7$, which is forbidden. Thus no two vertices of $A$ are nonadjacent, and $A$ is a clique. The Petersen graph is triangle-free, so a clique has size at most two. If $|A| = 2$, the two vertices are adjacent. The attachments therefore occupy a single vertex or the two ends of one edge. These are the classes in $\mathcal{P}(n)$.

If $Q = F_2$, then $\eta(Q) = 5$ and $h = 5 + 2t$ is odd, so the deficit is $5 + n - h$. For odd $n$ this equals $\delta(n) = 5$ if and only if $n = h$ and $n \geq 15$, in which case $G$ is $F_2$. For even $n$ the orders $n$ and $h$ have opposite parity, so $n - h \geq 1$ and the deficit is at least 6. Equality with $\delta(n)$ holds if and only if $n - h = 1$ and $\delta(n) = 6$, hence $n \geq 16$ is even and $G$ is $F_2$ of order $n - 1$ with a single leaf. Here $t = (n - 6)/2 \geq 5$.

If $Q = F_0'$, then $\eta(Q) = 6$ and the deficit is $6 + n - h \geq 6$, with equality if and only if $n = h$. Then $\delta(n) = 6$, so $n \geq 16$ is even and $G$ is $F_0'$.

It remains to decide which vertices of $F_2$ may carry the leaf. Adding a leaf at a vertex $v$ of a $P_7$-free graph preserves $P_7$-freeness if and only if $v$ is not an end of an induced $P_6$: the leaf has degree one, so it is an end of every induced path that contains it, and the remainder is an induced path starting at $v$. Label $F_2$ as in the description before Lemma 5.2. The vertex set consists of the hubs $v_1$ and $v_3$, the midpoint $v_2$ of the unique 2-edge branch $v_1v_2v_3$, and the internal vertices of the long branches. The long branches are the $t$ matching paths through $A_1 \cup A_3$ and the path $v_1v_5v_4v_3$. The map that swaps $v_1$ with $v_3$, swaps $v_5$ with $v_4$, fixes $v_2$, and swaps the two ends of each matching edge preserves the cycle, the matching and every spoke, so it is an automorphism. It exchanges the two hubs.

No induced $P_6$ starts at a hub. Let $P$ be an induced path starting at $v_1$. If the second vertex is $v_2$, the third is $v_3$, and the fourth is either $v_4$ or a vertex of $A_3$. In the first subcase the only continuation is $v_5$, and $v_1v_5$ is a chord, so the induced path has order at most four. In the second the only continuation is the match in $A_1$, and that match is adjacent to $v_1$, so again the order is at most four. If instead the path leaves $v_1$ along a long branch, it may continue to $v_3$ and then to the $v_3$-end of one further long branch. The resulting path has order five. Its last vertex has no neighbour off that path except its partner on the same long branch, and that partner is adjacent to $v_1$. The path does not extend, and the edge $v_1v_2$ is a chord of any continuation through $v_2$. Thus no induced $P_6$ starts at $v_1$. The automorphism gives the same conclusion at $v_3$.

No induced $P_6$ starts at $v_2$. Every induced path that starts at $v_2$ has second vertex $v_1$ or $v_3$. From $v_1$ it can enter only one long branch, and the far end of that branch is $v_3$, which is adjacent to $v_2$. The path stops before $v_3$ and has order at most four.

Every other vertex is an end of an induced $P_6$. The remaining vertices are the internal vertices of the long branches, namely $A_1 \cup A_3 \cup \{v_4, v_5\}$. Let $p$ be such a vertex and adjacent to $v_1$, so $p \in A_1$ or $p = v_5$. The automorphism carries these vertices to the vertices adjacent to $v_3$. Since $t \geq 5$, at least two long branches avoid $p$. Choose two matching branches $v_1abv_3$ and $v_1cdv_3$ among them, with $a, c \in A_1$ and with $b, d \in A_3$ their matches, and consider the path $p, v_1, a, b, v_3, d$. The edges $pv_1, v_1a, ab, bv_3$ and $v_3d$ are present by construction. There is no chord. A vertex of $A_1$ is adjacent only to $v_1$ and its match, a vertex of $A_3$ is adjacent only to $v_3$ and its match, $A_1$ and $A_3$ are stable sets, and the hubs are nonadjacent. If $p \in A_1$, its match is neither $b$ nor $d$, so $p$ meets the path only at $v_1$. If $p = v_5$, the only neighbours of $v_5$ are $v_1$ and $v_4$, and $v_4$ is not on the path, so again $p$ meets the path only at $v_1$. The vertex $a$ meets the path only at $v_1$ and $b$, and $b$ meets the path only at $a$ and $v_3$. Thus the path is an induced $P_6$, and a leaf at $p$ extends it to an induced $P_7$.

A leaf at a hub or at $v_2$ therefore keeps the graph $P_7$-free, and a leaf at any other vertex does not. These are the only legal positions. The two hubs are equivalent under the automorphism, while $v_2$ is the unique common neighbour of the hubs, so the two placements are not isomorphic. In $F_2$ the hub $v_1$ has degree $2 + t$, because it is adjacent to $v_2$, to $v_5$ and to the $t$ vertices of $A_1$, and $v_2$ has degree 2. After the leaf is added, the neighbour of the leaf has degree $t + 3$ or degree 3 according as it is a hub or $v_2$. For $t \geq 5$ these degrees differ.

The graphs that meet $\delta(n)$ are pairwise nonisomorphic. Orders 15 and 16 are precisely the orders at which a Petersen-core extremizer and an extremizer with a different core coexist. A graph in $\mathcal{P}(n)$ has $n-10$ vertices of degree one, and the degrees of the core vertices adjacent to a leaf are $3+a$ and, when $b \geq 1$, $3+b$. The graph $F_2$ has minimum degree two. The graph $F_0'$ has minimum degree two. Each one-leaf extension of $F_2$ has exactly one vertex of degree one, and the two extensions are separated by the degree of its neighbour.

For $10 \leq n \leq 14$ one has $\delta(n) = n-10$, so only $\mathcal{P}(n)$ occurs, and the pairs $(a,b)$ are separated by the degrees $3+a$ and $3+b$. For $n = 15$ one has $\delta(15) = \min\{5,5\} = 5$. The class $\mathcal{P}(15)$ consists of the three pairs $(5,0)$, $(4,1)$ and $(3,2)$. Each graph has five vertices of degree one, and the degrees of the core vertices adjacent to those leaves are $\{8\}$, $\{7,4\}$ and $\{6,5\}$ respectively, so the three graphs are nonisomorphic. The graph $F_2$ of order 15 has $t = 5$ and minimum degree two, hence no vertex of degree one, and it is not isomorphic to any of the three. These are four classes. For $n = 16$ one has $\delta(16) = \min\{6,6\} = 6$. The class $\mathcal{P}(16)$ consists of the four pairs $(6,0)$, $(5,1)$, $(4,2)$ and $(3,3)$. Each graph has six vertices of degree one, and the corresponding core degrees are $\{9\}$, $\{8,4\}$, $\{7,5\}$ and $\{6,6\}$, so the four graphs are nonisomorphic. The graph $F_0'$ has no vertex of degree one. Each one-leaf extension of the order-15 graph $F_2$ has exactly one vertex of degree one; since $t = 5$, the neighbour of that leaf has degree 8 or degree 3. A graph with six leaves, a graph with no leaf, and a graph with one leaf are nonisomorphic, and the two one-leaf extensions are separated by the degree of the neighbour of the leaf. These are seven classes. For odd $n \geq 17$ only $F_2$ meets $\delta(n)$. For even $n \geq 18$ the three graphs $F_0'$ and the two one-leaf extensions are the only ones, and the same degree count separates them: $F_0'$ has no leaf, and the neighbour of the unique leaf has degree $t+3 = n/2$ or degree 3. □

An enumeration of the equality graphs of orders 10 through 18 finds 1, 1, 2, 2, 3, 4, 7, 1, 3 isomorphism classes. The count agrees with Theorem 5.6 and is not used in the proof.

**Lemma 5.7.** *Let $G$ be a connected strong $C_4$-free $P_7$-free graph that contains a triangle, satisfies $L(v) \geq 4$ for every vertex, and contains no two adjacent vertices of degree two. Then $G$ is isomorphic to the graph $H_{3,2}$ obtained by partitioning the vertices of $K_5$ into parts of sizes three and two and subdividing each of the six edges with one end in each part exactly once. This graph has*

$$(n, e, T, S) = (11, 16, 1, 29).$$

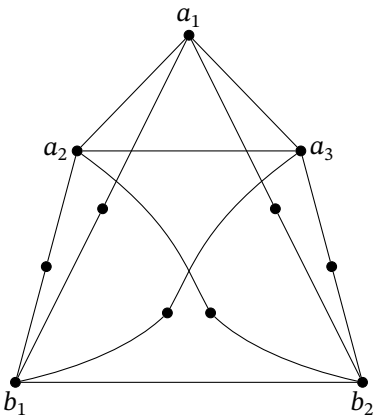



**FIGURE 4** | The graph $H_{3,2}$ of Lemma 5.7. The part $\{a_1, a_2, a_3\}$ spans the unique triangle, the part $\{b_1, b_2\}$ spans the remaining edge of $K_5$, and each of the six edges of $K_5$ between the parts is subdivided once. The two curved paths are the subdivisions of $a_2b_2$ and $a_3b_1$; they do not meet.

*Proof.* Let $A$ be the part of size three, let $B$ be the part of size two, and let $Z$ be the set of six subdivision vertices. The construction puts a triangle on $A$, an edge on $B$, no edge between $A$ and $B$, and exactly one vertex of $Z$ on each pair with one end in $A$ and one end in $B$. Thus $n = 11$ and $e = 3+1+12 = 16$. A vertex of $Z$ is adjacent to one vertex of $A$ and one vertex of $B$, which are nonadjacent, and no vertex of $Z$ lies in a triangle with an edge of $A$ or with the edge of $B$. The only triangle is the triangle on $A$, so $T = 1$ and $S = 2e - 3T = 29$. Each vertex of $A$ has degree four and lies in one triangle, so its deletion loss is 5. Each vertex of $B$ has degree four and lies in no triangle, so its deletion loss is 8. Each vertex of $Z$ has degree two, with nonadjacent neighbours, so its deletion loss is 4. No two vertices of $Z$ are adjacent. Figure 4 draws the graph.

There is no 4-cycle. The subgraph induced by $A \cup B$ is a triangle together with a disjoint edge. A cycle through a vertex $s \in Z$ uses both edges at $s$, so it contains the path from its neighbour in $A$ to its neighbour in $B$. Those two vertices have no common neighbour other than $s$.
There is no induced $P_7$. The set $A$ is a clique and the set $B$ is a clique, so an induced path contains at most two vertices of $A$ and at most two vertices of $B$. An induced path on seven vertices would therefore contain at least three vertices of $Z$. A vertex of $Z$ on an induced path is an end of the path, or else lies between its two neighbours. Suppose two vertices of $Z$ are internal. If their neighbours in $B$ are distinct, those two vertices of $B$ are adjacent, hence consecutive on the path, and the path contains a subpath $a\,s\,b\,b'\,s'\,a'$ with $a, a' \in A$. If their neighbours in $B$ coincide, the path contains a subpath $a\,s\,b\,s'\,a'$ with $a, a' \in A$. In both cases the edge $aa'$ is a chord. Thus at most one vertex of $Z$ is internal. At least three vertices of $Z$ lie on the path, and a path has only two ends, so the path contains exactly three vertices of $Z$: the two ends and one internal vertex. The other four vertices lie in $A \cup B$.
The internal vertex contributes a subpath $asb$ with $b \in B$. Suppose both vertices of $B$ lie on the path. They are consecutive, so the path contains the block $asbb'$. The fourth vertex of $A \cup B$ on the path is a vertex $a' \in A$, and $asbb'$ is a block of four consecutive vertices. There is no edge between $A$ and $B$, so $a'$ is not adjacent to $b'$. The only remaining possibility is that $a'$ is adjacent to $a$ on the path. The two ends of the resulting seven-vertex path lie in $Z$. The end adjacent to $a'$ has its other neighbour in $B$, because its neighbour on the path lies in $A$. Both vertices of $B$ are already on the path and neither is the successor of that end, so the other neighbour is a chord. Suppose instead that only one vertex of $B$ lies on the path. Then all three vertices of $A$ lie on the path. Those three vertices form a clique, so they do not lie on an induced path. This contradiction shows that $H_{3,2}$ is strong $C_4$-free and $P_7$-free, and the local hypotheses have already been checked. Thus $H_{3,2}$ is a graph of the kind named in the lemma.
It remains to show that no other isomorphism type occurs. Let $G$ satisfy the hypotheses and fix an induced triangle as the starting subgraph, still called $H$ during the growth. A labelled induced subgraph is enlarged by the following rule. If some vertex $v$ of the current graph $H$ has deletion loss $L_H(v) < 4$, computed in $H$, and $v$ has no neighbour in $G$ outside $H$, then the degree and the triangle count of $v$ have attained their values in $G$, so $L_G(v) < 4$. If $a$ and $b$ are adjacent vertices of degree two in $H$, then at least one of them has a neighbour in $G$ outside $H$. The rule then selects the pair $X = \{a, b\}$. It does not designate one endpoint of the pair in advance. In the deficient-vertex situation the rule selects the singleton consisting of that vertex. In either situation a new vertex required by the rule has neighbourhood $U$ in the current vertex set satisfying $U \cap X \neq \varnothing$. When $a$ and $b$ are the adjacent degree-two vertices, this is the condition $U \cap \{a, b\} \neq \varnothing$. If $H$ already satisfies $L_H \geq 4$ at every vertex and has no two adjacent vertices of degree two, its isomorphism type is compared with $H_{3,2}$. If that graph is still a proper induced subgraph of $G$, connectivity supplies a vertex outside $H$ with a neighbour in $H$, and every nonempty neighbourhood is admitted at that node.
For the next labelled vertex $z$, form every set $U \subseteq V(H)$ that meets the required set $X$, or every nonempty $U$ at an unrestricted node, and join $z$ precisely to $U$. Keep the child when the new graph has neither a 4-cycle subgraph nor an induced $P_7$. A new 4-cycle exists precisely when two vertices of $U$ have a common neighbour in the current graph. An induced $P_7$ created by the addition uses $z$ and six vertices of the current graph. Fix any embedding of the current graph into $G$. The neighbourhood, in the current vertex set, of a vertex that the rule requires is one of these sets $U$, and the child it determines is kept. An embedded proper prefix therefore extends. Every graph satisfying the hypotheses arises along some branch, with no restriction on its order.
The search tree is finite. Section A of the supplement records the audit. The same certificate is checked by the submission program `certificates/gen3/verify_gen3.py`. From the repository root the command is `python3 certificates/gen3/verify_gen3.py`. The tree has 169 labelled nodes and 52 leaves. By order 3, … , 17 the numbers of visited nodes are

$$1, 1, 1, 1, 2, 3, 5, 14, 23, 24, 23, 19, 24, 20, 8.$$

Four nodes are unrestricted. All four have order 11, and each is the graph $H_{3,2}$. The largest order reached is 17. That bound is an output of the search: the verifier imposes no order cutoff and performs no isomorphism pruning. A node at which a local condition fails is not discarded. The expansion continues, and every new neighbourhood is required to meet the deficient vertex or the adjacent pair of degree-two vertices. Every such expansion terminates, and it does so before any node other than the four copies of $H_{3,2}$ satisfies both local conditions. Every one-vertex extension of $H_{3,2}$ leaves the class or violates a local hypothesis. Therefore $G \cong H_{3,2}$. □

**Theorem 5.8.** *Every connected strong $C_4$-free $P_7$-free graph $G$ satisfies $S(G) \leq 3n$, with equality if and only if $G$ is the Petersen graph.*

*Proof.* The triangle-free case, including the equality, is Theorem 5.5. Suppose $G$ contains a triangle and $S(G) \geq 3n$, and choose such a connected graph of minimum order. Every smaller connected graph in the class satisfies $S \leq 3|V|$, and the inequality is strict whenever the smaller graph contains a triangle.

Suppose $s$ is a cut vertex. Induced paths that start at $s$ in distinct components of $G - s$ concatenate. Since $p(G) \leq 6$, some component determines a side $G_1$, induced by that component together with $s$, in which every induced path starting at $s$ has order at most three. Let $G_2$ be the subgraph induced by the remaining vertices together with $s$, and write $n_i = |V(G_i)|$. Then $n = n_1 + n_2 - 1$ and $n_1 \geq 2$. Lemma 5.3 gives $S(G_1) \leq 2(n_1 - 1)$. The graph $G_2$ is a smaller connected graph in the class. If it contains a triangle, minimality gives $S(G_2) < 3n_2$; if it does not, Theorem 5.5 gives $S(G_2) \leq 3n_2$. Lemma 2.3 therefore yields

$$S(G) \leq 2(n_1 - 1) + 3n_2 = 3n - (n_1 - 1) < 3n.$$

Thus $G$ has no cut vertex, and $G - v$ is connected for every vertex $v$.
Suppose $L(v) \leq 3$ for some vertex $v$. Then $S(G - v) = S(G) - L(v) \geq 3n - 3 = 3(n - 1)$. The graph $G - v$ is connected and lies in the class. It contains no triangle: otherwise minimality would give the strict inequality $S(G - v) < 3(n - 1)$. Theorem 5.5 therefore forces $G - v$ to be the Petersen graph, so $n = 11$ and $S(G - v) = 30$. Any two nonadjacent vertices of the Petersen graph have a common neighbour, so the neighbours of $v$ form a clique. The Petersen graph is triangle-free, and a clique in it has size at most two. The graph $G$ contains a triangle, so $v$ is adjacent to both ends of an edge and to no further vertex of $G - v$. Then $d(v) = 2$, $t(v) = 1$ and $L(v) = 2 \cdot 2 - 3 \cdot 1 = 1$, hence $S(G) = 31 < 33$. This contradiction shows that $L(v) \geq 4$ for every vertex $v$.
Suppose $u$ and $w$ are adjacent vertices of degree two. The bound on the deletion loss forces $t(u) = t(w) = 0$, so neither vertex lies in a triangle and the two neighbours of $u$ are nonadjacent. Let $x$ be the neighbour of $u$ other than $w$, and let $y$ be the neighbour of $w$ other than $u$. Then $x \neq y$, the edge $xy$ is absent, and $x$ is not adjacent to $w$. Deleting $\{u, w\}$ removes the three edges $xu$, $uw$ and $wy$ and preserves every triangle. The graph that remains may be disconnected. Its number of edges is $e(G) - 3$ and its number of triangles is $T(G)$, so its value of $S$ is $S(G) - 6 \geq 3n - 6 = 3(n - 2)$. For a disconnected graph, $S$ is the sum of the values on the components. Each component is a smaller connected graph in the class. The triangle of $G$ survives in one component, and minimality makes the inequality strict for that component, while every other component contributes at most three times its order. The sum is therefore strictly less than $3(n - 2)$, a contradiction.
Thus $G$ satisfies the hypotheses of Lemma 5.7. That lemma gives $n = 11$ and $S(G) = 29 < 33$, which is impossible. No connected graph in the class with a triangle satisfies $S \geq 3n$. Combined with Theorem 5.5, this is the stated bound, and equality holds only for the Petersen graph. □

**Theorem 5.9** (Theorem A). *Let $G$ be connected and strong $C_4$-free with $n \geq 2$. Then $p(G) \geq r(G) + \ell(G)$ whenever $\ell(G) \leq 3$, whenever $r(G) \leq 2$ and $\ell(G) \leq 4$, whenever $r(G) = 4$ and $\ell(G) \leq 4$, and whenever $r(G) \geq 5$ and $\ell(G) \leq 4$. If $r(G) = 4$ and $p(G) = 7$, then $\ell(G) \leq 2$.*

*Proof.* Write $r = r(G)$ and $c = \lceil \ell(G) \rceil$. The last sentence is Corollary 4.5. For the inequality we need $p \geq r + c$ in the listed ranges.

- $r = 1$: Lemma 3.1.
- $r = 2$, $c \leq 2$: Lemma 3.2.
- $r = 2$, $c = 3$: if $p \leq 4$, then Lemma 3.3 gives $\ell \leq 2$, a contradiction. So $p \geq 5$.
- $r = 2$, $c = 4$: if $p \leq 5$, then Theorem 3.6 gives $\ell \leq 3$, a contradiction. So $p \geq 6$.
- $r = 3$, $c \leq 2$: Theorem 2.4 gives $p \geq 5$.
- $r = 3$, $c = 3$: if $p \leq 5$, then by Lemma 3.5 and $r = 3$ we have $G = C_6$, with $\ell = 2$, a contradiction. So $p \geq 6$. Theorem 3.6 gives $\ell \leq 3$ for every $P_6$-free graph in the class, which does not rule out $c = 3$.
- $r \geq 4$, $c \leq 3$: Theorem 2.4 gives $p \geq 2r - 1 \geq r + 3$.
- $r = 4$ and $\ell \leq 4$: Theorem 2.4 gives $p \geq 7$. If $p = 7$, Corollary 4.5 gives $\ell \leq 2$, so $p \geq r + \ell$. If $p \geq 8$, Remark 4.7 gives the same inequality.
- $r \geq 5$ and $\ell \leq 4$: Theorem 2.4 gives $p \geq 2r - 1 \geq r + 4 \geq r + c$.

□

Equality in REAL-133 occurs for $C_5$ ($r = 2$, $\ell = 2$, $p = 4$), the Petersen graph ($r = 2$, $\ell = 3$, $p = 5$) and $C_6$ ($r = 3$, $\ell = 2$, $p = 5$). We do not characterise all equality cases.

**Corollary 5.10.** *Every connected graph $G$ on at least two vertices with $\ell(G) \leq 4$ satisfies Conjecture 1.1. The conjecture remains open for graphs with $\ell > 4$.*

*Proof.* If $G$ contains a 4-cycle, then $c_{C_4}(G) = 0$ and the conjecture reads $p(G) \geq r(G) + 1$. A shortest path between two vertices at distance $\operatorname{diam}(G) \geq r(G)$ is induced.
Suppose $G$ is strong $C_4$-free, $\ell(G) \leq 4$, and the conjecture fails. Theorem 5.9 leaves only the range $r(G) = 3$ and $3 < \ell(G) \leq 4$. A failure means $p(G) < r(G) + \ell(G) \leq 7$, so $p(G) \leq 6$. Theorem 5.8 then gives $\ell(G) \leq 3$, a contradiction.
The comparison does not reach $\ell > 4$. In that range Corollary 6.1 still leaves the statements $\mathrm{GEN}_K$ for $K \geq 4$ and $\mathrm{RAD}(r)$ for $r \geq 5$. □

*Remark 5.11.* Every connected graph $G$ with $\ell(G) < 5$ satisfies

$$p(G) \geq r(G) + \left\lfloor \ell(G)^{c_{C_4}(G)} \right\rfloor.$$

The exponent follows the convention of Conjecture 1.1: if $c_{C_4}(G) = 0$, then the power equals 1. If $G$ contains a 4-cycle, both the real form and this floor form reduce to $p \geq r + 1$, which follows from a geodesic. If $G$ is strong $C_4$-free and $\ell \leq 4$, Corollary 5.10 gives $p \geq r + \ell$, and $\ell \geq \lfloor \ell \rfloor$. Suppose $G$ is strong $C_4$-free and $4 \leq \ell < 5$, and suppose the floor form fails. Then $\lfloor \ell \rfloor = 4$, so $p \leq r + 3$. Theorem 2.4 gives $p \geq 2r - 1$, so $r \leq 4$. If $r \leq 3$, then $p \leq 6$, and Theorem 5.8 gives $\ell \leq 3$, contradicting $\ell \geq 4$. If $r = 4$, then $p = 7$, and Corollary 4.5 gives $\ell \leq 2$, again a contradiction. The real-valued conjecture for $\ell > 4$ remains open.

## 6 | The reduction above density four

For integers $K \geq 3$ and $r \geq 4$ consider the following statements.

$\mathrm{GEN}_K$: every connected strong $C_4$-free graph $G$ with $p(G) \leq 2K$ satisfies $S(G) \leq Kn$.

$\mathrm{RAD}(r)$: every connected strong $C_4$-free graph with $r(G) = r$ and $p(G) = 2r - 1$ satisfies $\ell(G) \leq r - 1$.

Corollary 4.5 is the case $r = 4$ of $\mathrm{RAD}(r)$, in the stronger form $\ell \leq 2$.

**Corollary 6.1.** *Let $G$ be a connected strong $C_4$-free graph that violates Conjecture 1.1. Then $\ell(G) > 4$. Put $K = \lceil \ell(G) \rceil - 1 \geq 4$. Then either*
*(a) $r(G) \leq K$ and $p(G) \leq r(G) + K \leq 2K$, so $G$ violates $\mathrm{GEN}_K$; or*
*(b) $r(G) = K + 1 \geq 5$ and $p(G) = 2r(G) - 1$, and $\ell(G) > r(G) - 1$, so $G$ violates $\mathrm{RAD}(K + 1)$.*

*Proof.* Write $r = r(G)$, $p = p(G)$ and $c = \lceil \ell(G) \rceil = K + 1$. Then $\ell > 4$ by Corollary 5.10, so $K \geq 4$. A violation means $p \leq r + c - 1 = r + K$. Theorem 2.4 gives $2r - 1 \leq p \leq r + K$, so $r \leq K + 1$. If $r \leq K$, then $p \leq r + K \leq 2K$, and $S > Kn$ because $\ell > K$. If $r = K + 1$, then $2r - 1 \leq p \leq 2K + 1 = 2r - 1$, and $\ell > K = r - 1$. The radius in (b) is at least 5. □

Since $\mathrm{GEN}_K$ ignores the radius, it is stronger than alternative (a). The statement $\mathrm{GEN}_K$ might fail for large $K$ even if REAL-133 holds; see Remark S2.5.

**Proposition 6.2.** *For $k \geq 3$ let $\theta_k$ consist of two vertices $x, y$ joined by $k$ internally disjoint paths $xa_ib_iy$. Then $\theta_k$ is strong $C_4$-free, $n = 2k + 2$, $r = 3$, $p = 6$ and $S = 6k = 3n - 6$. Thus $p - r - \ell = 6/n \to 0$.*

*Proof.* Every cycle consists of two complete branches and so has length 6; in particular there is no 4-cycle. There are no triangles, so $S = 2e = 6k$. Every vertex has eccentricity 3: $d(x, y) = 3$ and $d(a_i, b_j) = 3$ for $i \neq j$. The path $a_jxa_ib_iyb_l$ (with $j, i, l$ distinct) is induced. Conversely, let $P$ be an induced path. If $P$ avoids $x$ and $y$, it lies inside one edge $a_ib_i$. If $P$ contains $x$ but not $y$, then $x$ has at most two neighbours on $P$, each followed on $P$ by at most its $b$-vertex, so $|P| \leq 5$. If $P$ contains $x$ and $y$, then every $a_j$ on $P$ is adjacent on $P$ to $x$ and every $b_j$ to $y$, at most one branch lies entirely on $P$, and so $P$ consists of $x$, $y$, at most one full branch and at most one further neighbour of each of $x$ and $y$. Hence $|P| \leq 6$. □

**Corollary 6.3.** *Let $G$ be a connected strong $C_4$-free graph with $p(G) = 6$. Then $S(G) \leq 3n - 1$ and $\ell(G) < 3$. On each of the classes $(r, p) = (2, 6)$ and $(r, p) = (3, 6)$ the supremum of $\ell$ is 3, and the supremum is not attained.*

*Proof.* The graph is $P_7$-free. Theorem 5.8 gives $S(G) \leq 3n$, with equality only for the Petersen graph. The Petersen graph has induced-path order 5, so $S(G) \leq 3n - 1$ and $\ell(G) < 3$.
Both constructions below are strictly less than 3 and tend to 3 as the order tends to infinity. The supremum on each class is therefore 3, and it is not attained.
For the class $(2, 6)$, take the graph $F_2$ of Theorem 5.6 with parameter $t \geq 2$. In the description preceding Lemma 5.2 the hubs are $v_1$ and $v_3$, the unique branch of length 2 is $v_1v_2v_3$, and the remaining $v_1$–$v_3$ paths are the $t$ matching paths through $A_1 \cup A_3$ together with $v_1v_5v_4v_3$. These $t + 1 \geq 3$ paths are branches of length 3. The order is $n = 5 + 2t$. The graph

is triangle-free, and the table of Lemma 5.2 gives $2e = 3n - 5$, so $\ell = 3 - 5/n$. Every vertex other than the two hubs is adjacent to a hub, and $v_2$ is adjacent to both hubs, so every vertex is within distance 2 of $v_2$. A vertex of $A_1$ is not adjacent to $v_2$. Each hub has degree $t + 2 = (n-1)/2 < n - 1$, and every other vertex has degree 2, so no vertex dominates the graph. The radius is 2. The proof of Theorem 5.6 shows that every induced path has order at most 6. For the matching lower bound, choose three branches of length 3, written $v_1abv_3$, $v_1cdv_3$ and $v_1efv_3$. The path $a, v_1, c, d, v_3, f$ is induced: the hubs are nonadjacent, and an internal vertex of any one of these branches is adjacent only to its two neighbours on that branch. Thus $(r, p, \ell) = (2, 6, 3 - 5/n)$.
For the class $(3, 6)$, Proposition 6.2 gives the graph $\theta_k$ for every $k \geq 3$, with $\ell = 3 - 6/n$. □

The two families of Corollary 6.3 show that no proof of $\mathrm{GEN}_3$ can proceed by establishing a uniform positive margin below 3. Corollary 5.10 is REAL-133 for every connected graph with $\ell \leq 4$. Its proof uses Theorem 5.9 and Theorem 5.8. The range $\ell > 4$ remains open. Corollary 6.1 still leaves $\mathrm{GEN}_K$ for $K \geq 4$ and $\mathrm{RAD}(r)$ for $r \geq 5$.

## 7 | What is proved and what is open

The comparisons with Chen [3], Chen and Xu [5], Chen, Wu and Xu [4], Chudnovsky and Stacho [6], DeLaViña, Larson, Pepper and Waller [8], Karthick and Maffray [13], and Brandstädt and Hoàng [1] are in the introduction, in Section 5 and in Remarks 3.7 and 3.8. Table 1 records the results used for the range $\ell \leq 4$. An open row is not a hypothesis of a proved row. A finite search is a proof here only in the form fixed for Lemma 5.7, and for the two-triangle search of Lemma S5.3 in the supplement.
If $p = 6$, Corollary 6.3 gives $S \leq 3n - 1$ and $\ell < 3$. The pairs $(r, p) = (2, 6)$ and $(3, 6)$ are excluded from the range $\ell \geq 3$. The value 3 is a sharp supremum and is not attained. The graph $F_2$ of Theorem 5.6, once it has at least three branches of length 3, has radius 2 and $\ell = 3 - 5/n$, and the graph $\theta_k$ has radius 3 and $\ell = 3 - 6/n$.
Lemma S4.2 gives $\ell < 4$ for induced-path order 7 only under the triangle-free hypothesis. If $p = 7$ and $\ell \geq 4$, then $G$ contains a triangle. The condition does not forbid $(r, p) = (2, 7)$ in a triangle-free graph. The supplement gives a triangle-free example with parameters $(r, p, n, S, \ell) = (2, 7, 12, 34, 17/6)$, and it records the pairs that remain necessary in the bands $4 < \ell \leq 5$ and $5 < \ell \leq 6$.

| Statement | Status |
|---|---|
| $M(n)$ for connected strong $C_4$-free $P_6$-free graphs, with uniqueness for $n \geq 7$ | proved, Theorem 3.6 |
| graphs with $r = 4$ and $p = 7$, the formula $S = 2n - b - m$, and the equality family | proved, Theorem 4.3 and Corollaries 4.5, 4.6 |
| $2e \leq 3n$ for connected triangle-free strong $C_4$-free $P_7$-free graphs, equality only for the Petersen graph | proved, Theorem 5.5 |
| the exact value $f(n)$ and the extremal graphs, $n \geq 10$ | proved, Theorem 5.6 |
| $S \leq 3n$ for every connected strong $C_4$-free $P_7$-free graph, equality only for the Petersen graph | proved, Theorem 5.8 |
| if $p = 6$, then $S \leq 3n - 1$ and $\ell < 3$; the supremum 3 is sharp and unattained for $(2, 6)$ and for $(3, 6)$ | proved, Corollary 6.3 |
| REAL-133 for $\ell \leq 3$; for $r \leq 2$ and $\ell \leq 4$; for $r = 4$ and $\ell \leq 4$; for $r \geq 5$ and $\ell \leq 4$ | proved, Theorem 5.9 |
| REAL-133 for every connected graph with $\ell \leq 4$ | proved, Corollary 5.10 |
| REAL-133 for $\ell > 4$, and REAL-133 with no restriction on $\ell$ | open |

**TABLE 1** | Proved statements for $\ell \leq 4$, and what remains open.

The rooted bounds, the endpoint obstructions, the high-radius constructions, the one-triangle bound, and the edge-deficit estimates are in the supplement. The dependency note there records which of those arguments are hypotheses of the theorems. The certificate for Lemma 5.7 is used in that lemma. The cut-vertex step in Theorem 5.8 uses Lemmas 5.3 and 2.3. No structure theorem for connected strong $C_4$-free graphs of radius $r \geq 5$ and path order $2r - 1$ is proved here.

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